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Mock CAT - 18 2020 (New Pattern)

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VARC

LRDI

QA

Sec 1

Direction for questions (1-6): Read the given passage and answer the questions that follow.

The principle of Darwinian evolution now is not just an explanatory theory, but also a debunker of theism. As such, it has been elevated to a status of unquestionable truth to the extent that biologists who may have doubts on its fireproof status would not admit so in public in case they become pariahs in their community. The other downside of holding such an absolutist position is that any proposal that may hint at other mechanisms that do not comfortably fit with the orthodox beliefs are dismissed outright as they may imply some universal intelligence or teleological plan that smack of a heavenly planner. Its most enthusiastic adherents assert that the theory of evolution has no room for other mechanisms hence no other possibilities are conceivable or indeed allowable. This, in spite of the fact that an unresolved residue is always present in science even after the most successful application of reductionist principles with the corollary that all theories should be taken to be provisional and incomplete. This unscientific attitude burdens the Darwinian theory of evolution with a weight it just cannot carry.

There is no doubt that the theory of evolution is handsomely supported by the fossil records and has considerable explanatory powers. However, there are two areas where one finds the current version of evolution theory unconvincing. The first is the assertion that evolution is the sole mechanism that drives matter towards biological development. The second is its incapacity to explain the emergence of mentality.

Physicists tell us that following the big bang the only element in existence was hydrogen, the simplest in the periodic table. It was from these humble beginnings that the remaining ninety odd increasingly complex elements and their vastly more numerous and complex combinations were gradually synthesised as eons passed. This points to the existence of a natural law which is embedded in the nature of atomic physics and provides the potential of “evolution” of more complex elements from simpler ones.

The second difficulty one has is related to the spontaneous appearance of information carrying replicating systems and ultimately of what one may call mentality. At the pre-biotic stage of evolution, Darwinian competition cannot, by definition, assist the evolution process. Natural selection requires that primitive life is already there for the process to begin. The assumption is therefore made that “mindless and blind” unguided processes have spontaneously resulted in a self-replicating entity that encodes information, the precursor of the information carrying DNA. This process has been described by some as “evolution of the gaps” to rhyme with “God of the gaps.” It is at this stage that another explanation may help. Since ancient times, philosophers, scientists and particularly mystics held that matter and mentality are inextricably mixed. The mind is capable of directing the motion of the limbs, i.e. to control god-like “the motion of the atoms,” as Erwin Schrodinger observed. This complementary leverage of mind-over-matter and matter-over-mind is an extraordinary fact. The phenomenon of mentality, highly developed as consciousness in man, is also apparent in primates and in diminishing degree as one travels down the evolutionary tree of the animal kingdom. A bacterium can be said to possess a faint glimmer of mentality expressed in its ability to react intelligently to the environment. As the gradation of mentality is continuous, one can see no justification in stopping its diminution with a discontinuity when prions or ultimately “inorganic matter” is reached. The idea that mentality is just a stepwise epiphenomenon of complexity, usually illustrated by the hypothetical example of a computer which, if made large enough, will spontaneously show signs of mentality or consciousness is unproven and, in my view, absurd.

Q.1 [11594329]

According to the passage, which of the following can be inferred about “the evolution of the gaps”?

1 ○ It is the rational and scientific counterpart of the concept of “God of the gaps.”

2 ○ An unscientific explanation that scientists use to hide any scientific loopholes in a theory.

3 ○ The Darwinian theory of competition can't be utilized to understand evolution.

4 ○ It is insufficient to provide an explanation for the missing link in the process of evolution.

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Answer key/Solution

Solution:
Correct Answer : 4
Your Answer : 3
Refer to the lines, "The assumption is therefore made that "mindless and blind" unguided processes have spontaneously resulted in a self-replicating entity that encodes information, the precursor of the information carrying DNA. This process has been described by some as "evolution of the gaps" to rhyme with "God of the gaps."... It is at this stage that another explanation may help." So, the author clearly states that this evolution of the gaps is not enough. We need another theory. Words like "mindless and blind" suggest that the author's tone is not positive. The other options are either extreme or unsupported by the passage.

Bookmark FeedBack

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Q.2 [11594329]

According to the passage, the existence of a natural law:

1 ○ makes it difficult for the author to accept the findings of Darwin.

2 ○ proves that the findings of Darwin are not practically applicable.

3 ○ adds to the author's dilemma in accepting the theory of evolution.

4 ○ was the main reason for the evolution of complex elements from humble beginnings.

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Answer key/Solution

Solution:
Correct Answer : 3
Your Answer : 4
Refer to the line "The first is the assertion that evolution is the sole mechanism that drives matter towards biological development." So, the author has a problem with this belief and not with the existence of a natural law. In fact, the possibility of the existence of a natural law strengthens the author's scepticism about evolution being the sole mechanism as this law would have played a vital role in the process too. Hence, option 3 is a better answer than option 1.

Bookmark FeedBack

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Q.3 [11594329]

According to the passage, which of the following is not a reason for the author to find the current version of evolution theory unconvincing?

1 ○ The existence of fossil records

2 ○ The presumption of the absence of mentality in simple matter

3 ○ The elevated status of the principle of Darwinian evolution

4 ○ The possible existence of a natural law

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Answer key/Solution

Solution:
Correct Answer : 1
Your Answer : 4
Refer to the first line of the second paragraph. The author mentions fossil records as a reason to support the theory. Hence, this is the correct answer.

Bookmark FeedBack

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Q.4 [11594329]

In the context of this passage, which of the following is the best example of an “unscientific attitude”?

1 ○ A renowned critic insisting on a debate before considering that he was wrong in his evaluation of a movie.

2 ○ A batsman being skeptical about the suggestions of the new coach regarding his batting technique.

3 ○ A scholar of Theology rejecting the findings of an eminent but atheist scientist after thorough evaluation.

4 ○ A group of researchers refusing to consider data presented by a new scholar.

Answer key/Solution

Solution:
Correct Answer : 4
Refer to the second half of the first paragraph. The line "Its most enthusiastic adherents assert that the theory of evolution has no room for other mechanisms ... with the corollary that all theories should be taken to be provisional and incomplete...this unscientific attitude burdens the Darwinian..." So, according to the author, refusal to consider any new possibility is unscientific.

Option 1 – The movie critic might be right in refusing to accept any wrong judgement. The option is vague regarding the validity of the accusation.

Option 2 – The batsman might be right in his stance. The coach might be wrong.

Option 3 – The scholar of Theology might be right in not accepting the findings. The scientist might not have done good research work.

Option 4 – This is correct. Refusing to even consider any data is unscientific. The researcher may reject it after evaluation. But not considering it in the first place is, as per the passage, unscientific.

Bookmark FeedBack

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Q.5 [11594329]

Why does the author write the line “This complementary leverage of mind-over-matter and matter-over-mind is an extraordinary fact”?

1 ○ To marvel at the inherent complexity and incomprehensibility of the process of evolution

2 ○ To highlight a loophole in the current theory of evolution accepted by many

3 ○ To prove that Darwinian theory of evolution cannot explain the emergence of mentality

4 ○ To question the spontaneous appearance of mentality in the evolution process

Answer key/Solution

Solution:
Correct Answer : 2
Options 1 and 4 are too generic. The author is not questioning the entire process of evolution. Option 3 is irrelevant as the line in question is mentioned in a different context. Option 2 is the correct answer as the author writes this line to start a process of discussion.

Bookmark FeedBack

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Q.6 [11594329]

Which of the following is the primary focus of the author in this passage?

1 ○ The existence of logical problems in the explanation of theory of evolution

2 ○ The unscientific attitude demonstrated by many scientists when it comes to Darwin

3 ○ The God-like status of Darwin in the field of evolutionary biology

4 ○ The lack of explanation regarding the emergence of mentality in the process of evolution

✖

Answer key/Solution

Solution:
Correct Answer : 1
Your Answer : 4
Option 1 - This is the correct answer. The author mentions two problems and tries to analyse those.

Option 2 - This is too narrow and also distorted. "Many scientists" and the line is written in only one paragraph.

Option 3 - It doesn't match the tone or intention of the author.

Option 4 - It is too narrow. It is mentioned only towards the end of the passage.

Bookmark FeedBack

Direction for questions (7-12): Read the given passage and answer the questions that follow.

The next summer the truce for a year ended, after lasting until the Pythian Games. During the armistice the Athenians expelled the Delians from Delos, concluding that they must have been polluted by some old offence at the time of their consecration, and that this had been the omission in the previous purification of the island, which, as the writer has related, had been thought to have been duly accomplished by the removal of the graves of the dead. The Delians had Atramyttium in Asia given them by Pharnaces, and settled there as they were removed from Delos.

Meanwhile Cleon prevailed on the Athenians to let him set sail at the expiration of the armistice for the towns in the direction of Thrace with twelve hundred heavy infantry and three hundred horses from Athens, a large force of the allies, and thirty ships. The Athenians from the ships had already taken Torone, and their land forces following at his heels burst in with him with a rush over the part of the old wall that had been pulled down, killing some of the Peloponnesians and Toronaeans in the melee, and making prisoners of the rest, and Pasitelidas their commander amongst them. Brasidas meanwhile had advanced to relieve Torone, and had only about four miles more to go when he heard of its fall on the road, and turned back again. Cleon and the Athenians set up two trophies, one by the harbour, the other by the fortification and, making slaves of the wives and children of the Toronaeans, sent the men with the Peloponnesians and any Chalcidians that were there, to the number of seven hundred, to Athens; whence, however, they all came home afterwards, the Peloponnesians on the conclusion of peace, and the rest by being exchanged against other prisoners with the Olynthians. About the same time Panactum, a fortress on the Athenian border, was taken by treachery by the Boeotians. Meanwhile Cleon, after placing a garrison in Torone, weighed anchor and sailed around Athos on his way to Amphipolis.

About the same time Phaeax, son of Erasistratus, set sail with two colleagues as ambassador from Athens to Italy and Sicily. The Leontines, upon the departure of the Athenians from Sicily after the pacification, had placed a number of new citizens upon the roll, and the commons had a design for redividing the land; but the upper classes, aware of their intention, called in the Syracusans and expelled the commons. These last were scattered in various directions; but the upper classes came to an agreement with the Syracusans, abandoned and laid waste their city, and went and lived at Syracuse, where they were made citizens. Afterwards some of them were dissatisfied, and leaving Syracuse occupied Phocaeae, a quarter of the town of Leontini, and Bricinniae, a strong place in the Leontine country, and being there joined by most of the exiled commons carried on war from the fortifications.

The Athenians hearing this, sent Phaeax to see if they could not by some means so convince their allies there and the rest of the Sicilians of the ambitious designs of Syracuse as to induce them to form a general coalition against her, and thus save the commons of Leontini. Arrived in Sicily, Phaeax succeeded at Camarina and Agrigentum, but meeting with a repulse at Gela did not go on to the rest, as he saw that he should not succeed with them, but returned through the country of the Sicels to Catana, and after visiting Bricinniae as he passed, and encouraging its inhabitants, sailed back to Athens.

Q.7 [11594329]

During the armistice, according to the author, why did the Athenians expel the Delians from Delos?

1 ○ They were charged with polluting the warfare.

2 ○ The offence of removing the graves made the Athenians expel the Delians.

3 ○ They were believed to have been polluted by some old offence.

4 ○ None of the above.

Answer key/Solution

Solution:
Correct Answer : 3
Options (1), (2) and (3) may seem correct but option (3) is more appropriate an answer as it actually gives a reason. Option (1) is incomplete. Option (2) distorts the information provided. Refer to the line: "…had been thought to have been duly accomplished by the removal of the graves of the dead."

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Q.8 [11594329]

All of the following sentences are true except:

1 ○ the upper class negotiated with the Syracusans in expelling commoners from the state.

2 ○ Phaeax was sent to save the commons of Leontini by waging a war with Gela.

3 ○ a fortress on the Athenian border was taken by the Boeotians.

4 ○ the truce of the war lasted till the Pythian Games.

Solution:

Answer key/Solution

Correct Answer : 2

According to the given passage, options (1), (3) and (4) are true. Option (2) is not true because Phaeax was sent to talk with the other Peloponnesian states to form a coalition where he was repulsed by Gela. Hence, option (2) is the answer.

Bookmark FeedBack

Direction for questions (7-12): Read the given passage and answer the questions that follow.

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Q.9 [11594329]

According to the 1st paragraph of the given passage, which of the following defines ‘consecration’?

1 ○ The action of commanding someone

2 ○ The action of ordaining someone

3 ○ The action of seducing someone

4 ○ The action of following someone

Answer key/Solution

Solution:
Correct Answer : 2
The correct definition of the word would be ordaining someone and not commanding someone although both are quite similar to each other. Since a war was about to begin and the orders given by the Athenians to the Delians were official orders, option (2) is the best answer. Options (3) and (4) are incorrect.

Bookmark FeedBack

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Q.10 [11594329]

What can be derived from the said journey of Phaeax, keeping in mind the last sentence of the passage?

1 ○ It was an unsuccessful one because he was nauseated back at Gela.

2 ○ He was successful in spreading the word of the Athenians in all the states.

3 ○ He sailed back to Athens after working to persuade the states to form a coalition against Syracuse.

4 ○ All of the above.

Answer key/Solution

Solution:
Correct Answer : 3
In option (1),"nauseated back" is not something that can be concretely derived. In option (2), "all the states" can't be inferred. Option (3) is very specific. Refer to the line “The Athenians hearing this, sent Phaeax to see if they could not by some means so convince their allies there and the rest of the Sicilians of the ambitious designs of Syracuse as to induce them to form a general coalition against her…” and the last line of the passage also states “…as he passed, and encouraging its inhabitants, sailed back to Athens.” Hence, option (3) is the correct answer.

Bookmark FeedBack

Direction for questions (7-12): Read the given passage and answer the questions that follow.

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Q.11 [11594329]

In this passage, based on the tactics adopted by the Athenians, which of the following can be inferred?

1 ○ War can come to an end via negotiation.

2 ○ War beyond its impact - there is a way via which negotiation can happen.

3 ○ Negotiation was a rare phenomenon in early warfare .

4 ○ Coalition was the basic approach of the Athenians.

Answer key/Solution

Solution:
Correct Answer : 4
Option (1) is far fetched. The passage is silent on negotiation. Option (2) is also an extreme conclusion. In option (3), "rare" can't be determined. The last paragraph talks about Athenians' attempts to form a coalition against Syracuse. Therefore, option (4) is correct.

Bookmark FeedBack

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Q.12 [11594329]

What is the narrative structure of the given passage?

1 ○ The point of approach is from the third person point of view.

2 ○ A commentary on war where the author is involved.

3 ○ A biography of a war, so the narrative structure is biographical.

4 ○ All of the above.

Answer key/Solution

Solution:
Correct Answer : 3
As the question asks about the narrative structure of the passage, we should look into the technical aspect of it. Option (1) talks about the narrator and not the overall narrative pattern. It is also distorted as it creates a wrong causal relationship between war narrative and its pattern. Option (2) - One line can't determine the narrative pattern of the passage. Option (2) is confined only to "…which, as the writer has related, had been thought to have been duly accomplished by the removal of the graves of the dead". Option (3) is the best option as the passage talks about the biography of a particular era.

Bookmark FeedBack

Direction for questions (13-18): Read the given passage and answer the questions that follow.

Can you identify what or whom the following describes: is widely disliked around the world; might have been ditched by some supporters earlier had convincing alternatives existed; has had a difficult six months; and refuses to go quietly? Here's another clue: this is not a column about politics. The answer is the dollar. It is the most unloved of major currencies, apart from all the others. And, oddly, it has been given a fillip by a messy election result at home.

Or perhaps that is not so odd. The dollar's resilience has been one of the more monotonous motifs in financial markets in recent years. Dollar strength is twinned with another hardy theme—the growing heft of America's companies, notably its tech giants, in global equity markets. The dollar matters for America, but it matters for everywhere else, too. A weaker dollar would trigger a period of catch-up by the rest of the world's economies and asset markets. Such a prospect is seemingly delayed.

A reason for dollar resilience is growing doubts over fiscal stimulus in America. The election was supposed to be the start of a new era of fiscal largesse. Agreeing on any kind of policy now looks hard. If Joe Biden enters the Oval Office in January, he is likely to face a divided Congress. But dollar strength is not solely down to politics. It is as much about the economic consequences of a resurgent coronavirus as it is about dashed hopes of a blue-wave election.

Begin with the blue wave that didn't crest. Before the election, an idea had taken hold, fuelled by pollsters and election forecasters, that a clean sweep of the White House and both houses of Congress by the Democratic Party was highly likely. The upshot would be a weaker dollar.

In 2016 a similar prospect of fiscal easing drove the dollar up, not down. This needs some explaining. The difference is that four years ago, the Federal Reserve was expected to offset the stimulative effect of tax cuts by raising interest rates in order to contain inflation—thus supporting the dollar. But with the economy now weak, the Fed has committed itself to easy money. A fiscal-stimulus package would be an unimpeded spur to aggregate demand, leading to more imports, a wider trade deficit and a weaker dollar. And a weaker dollar would in turn help the rest of the world, partly because of its role as a borrowing currency beyond America's shores. A lot of emerging-market companies and governments have dollar debts, so a weaker greenback acts as an indirect stimulus to global growth.

Instead, the dollar has perked up a bit. That is because the dollar is special in another way. Dollar assets, notably shares, are more prized when the outlook seems less certain. Holders of dollars cling on to them for longer, rather than swap them for other currencies. This goes for wealthy savers in emerging economies, or Chinese or South Korean exporters, say, who have earned dollars on sales.

It also goes for institutional investors at home who might have thought of cashing in some of their expensive-looking American tech stocks for a wager on cheap-looking cyclical stocks in Europe or Asia. The diminished prospect of fiscal stimulus is one reason why this "reflation trade" is less alluring. The resurgence of coronavirus infections is another. Much of Europe is now in soft lockdown. Its economy is losing steam. The appeal of cyclical stocks is similarly ebbing. Investors have instead piled back into tech firms, which benefit from the stay-at-home economy. […]

Q.13 [11594329]

Which of the following is NOT mentioned as a reason behind the resilience of dollar?

1 ○ Ruined hopes of a blue-wave election in the US

2 ○ Doubts over fiscal stimulus in the US

3 ○ Economic consequences of Coronavirus

4 ○ None of the above

✔

Solution:
Correct Answer : 4
Your Answer : 4
Refer Para 3. All 1, 2, and 3 have been mentioned there.

Answer key/Solution

Bookmark FeedBack

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Q.14 [11594329]

What can be said most aptly about the opening paragraph of the passage?

1 ○ The author uses a question and a clue to engage the reader's attention.

2 ○ The author builds mystery around a topic and helps the reader eliminate an option.

3 ○ The author uses allusion to capture interest and introduce the discussion topic.

4 ○ The author uses a conversational tone and riddle-like diction to introduce a topic.

✖

Answer key/Solution

Solution:
Correct Answer : 3
Your Answer : 1
The writer makes an allusion to Donald Trump. This is made clear by the clue that "the column is not about politics", because the writer is aware that many readers would come to that conclusion because of her allusion. 3 is thus a better answer than 4, despite the fact that a conversational tone and a riddle-like diction are present. 1 and 2 are eliminated as they do not talk about introducing the topic.

Bookmark FeedBack

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Q.15 [11594329]

What does the author imply by the phrase "the blue wave that didn't crest" in Para 4?

1 ○ The Democratic party's win didn't lead to the weakening of the dollar.

2 ○ The Democratic party was unable to win at the large scale anticipated.

3 ○ The Democratic party performed opposite to expectations from it.

4 ○ Despite the Democratic Party's win, the Congress is divided on Biden's policies.

Answer key/Solution

Solution:
Correct Answer : 2
Crest refers to the peak of a wave. Para 4 later talks about expectations of a clean sweep (crest) by the Democratic Party (blue wave) that weren't fulfilled. So 2. 3 is incorrect as the Democratic party did win. 4 is eliminated as this statement is about election victory and not policies. 1 is not the implied meaning.

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Or perhaps that is not so odd. The dollar's resilience has been one of the more monotonous motifs in financial markets in recent years. Dollar strength is twinned with another hardy theme—the growing heft of America's companies, notably its tech giants, in global equity markets. The dollar matters for America, but it matters for everywhere else, too. A weaker dollar would trigger a period of catch-up by the rest of the world's economies and asset markets. Such a prospect is seemingly delayed.

A reason for dollar resilience is growing doubts over fiscal stimulus in America. The election was supposed to be the start of a new era of fiscal largesse. Agreeing on any kind of policy now looks hard. If Joe Biden enters the Oval Office in January, he is likely to face a divided Congress. But dollar strength is not solely down to politics. It is as much about the economic consequences of a resurgent coronavirus as it is about dashed hopes of a blue-wave election.

Begin with the blue wave that didn't crest. Before the election, an idea had taken hold, fuelled by pollsters and election forecasters, that a clean sweep of the White House and both houses of Congress by the Democratic Party was highly likely. The upshot would be a weaker dollar.

In 2016 a similar prospect of fiscal easing drove the dollar up, not down. This needs some explaining. The difference is that four years ago, the Federal Reserve was expected to offset the stimulative effect of tax cuts by raising interest rates in order to contain inflation—thus supporting the dollar. But with the economy now weak, the Fed has committed itself to easy money. A fiscal-stimulus package would be an unimpeded spur to aggregate demand, leading to more imports, a wider trade deficit and a weaker dollar. And a weaker dollar would in turn help the rest of the world, partly because of its role as a borrowing currency beyond America's shores. A lot of emerging-market companies and governments have dollar debts, so a weaker greenback acts as an indirect stimulus to global growth.

Instead, the dollar has perked up a bit. That is because the dollar is special in another way. Dollar assets, notably shares, are more prized when the outlook seems less certain. Holders of dollars cling on to them for longer, rather than swap them for other currencies. This goes for wealthy savers in emerging economies, or Chinese or South Korean exporters, say, who have earned dollars on sales.

It also goes for institutional investors at home who might have thought of cashing in some of their expensive-looking American tech stocks for a wager on cheap-looking cyclical stocks in Europe or Asia. The diminished prospect of fiscal stimulus is one reason why this "reflation trade" is less alluring. The resurgence of coronavirus infections is another. Much of Europe is now in soft lockdown. Its economy is losing steam. The appeal of cyclical stocks is similarly ebbing. Investors have instead piled back into tech firms, which benefit from the stay-at-home economy. […]

Q.16 [11594329]

Fill in the blanks in the correct order.

As per the last paragraph, _____ and _____ are the reasons why _____ and _____ is less alluring.

1 ○ Doubt over fiscal stimulus, Covid revival, buying cyclical stocks, selling US tech stocks

2 ○ European lockdown, doubt over fiscal stimulus, selling cyclical stocks, buying tech stocks

3 ○ A strong dollar, Covid revival, buying cyclical stocks, selling US tech stocks

4 ○ A divided Congress, Covid revival in Europe, buying European or Asian cyclical stocks, selling US tech stocks

Solution:

Correct Answer : 1

Refer the last para. Diminished prospects of a fiscal stimulus and resurgence of Coronavirus are given as reasons behind reflation trade, which is selling US tech stocks and buying cyclical stocks in Europe or Asia, less alluring. Hence, only 1 fits.

Answer key/Solution

Bookmark FeedBack

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Q.17 [11594329]

Which of the following can be inferred from Para 6 (the second last para)?

1 ○ Emerging economies such as China and South Korea value dollar more.

2 ○ The perception of safety associated with dollar makes it special.

3 ○ Wealthy savers and exporters are two groups that prefer dollar in uncertain times.

4 ○ Dollar can be called special because it has increased in value against expectations.

Answer key/Solution

Solution:
Correct Answer : 2
The para says that Dollar is special in another way, and as an explanation says how savers and exporters 'cling' to it, 'prize it more' when the outlook is uncertain.
This means they perceive it to be a safe currency. Hence, 2. 1 generalises a fact given about two groups for entire economies. 3 again generalises as the para refers to wealthy savers "from emerging economies". 4 inverts the cause-effect relationship. Dollar has increased in value because it is special in a way, and not vice versa.

Bookmark FeedBack

Direction for questions (13-18): Read the given passage and answer the questions that follow.

Can you identify what or whom the following describes: is widely disliked around the world; might have been ditched by some supporters earlier had convincing alternatives existed; has had a difficult six months; and refuses to go quietly? Here's another clue: this is not a column about politics. The answer is the dollar. It is the most unloved of major currencies, apart from all the others. And, oddly, it has been given a fillip by a messy election result at home.

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Q.18 [11594329]

The pattern of expected economic strategy of 2016 as explained in Para 5 is most similar to:

1 ○ The government invested in infrastructure and reduced interest rates, ultimately increasing job demand.

2 ○ The government subsidised the manufacturing of a product and advertised the product widely, ultimately increasing its price.

3 ○ Expecting low rainfall, the government increased the minimum support price for a crop and subsidised farm inputs, ultimately boosting its sowing.

4 ○ The government investing in the tourist attractions of a place, and promoting it as a brand, ultimately increasing the state tourism revenue.

Solution:
Correct Answer : 2
The pattern of the expected strategy is this:

Answer key/Solution

Tax cuts -> Increase in dollar supply -> Fall in dollar price

Interest rate increase -> Decrease in dollar demand -> Increase in dollar price Ultimate outcome- Support to dollar price

This pattern is mimicked by option 2.

Product subsidy -> Increase in product supply -> Fall in the price of the product

Advertisement -> Increase in product demand -> Increase in the price of the product

Ultimate outcome-> Increased price of the product

The others are eliminated because there are no opposing forces at play. In 1, investment in infra and reduction in interest rates, both result in increased job demand. In 3, the government MSP and farm input subsidy, both will induce the farmer to grow the crop. In 4 again, both investment in tourist attractions and their promotion will serve to boost tourism and thus increase state revenue from it.

Bookmark FeedBack

Directions for question (19): The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

Q.19 [11594329]

1. Nevertheless, the ideological differences between the left and the right are clearly articulated through their policies.
2. Broadly speaking, the left supports socialism, and the right can be said as associated with capitalism.
3. However, left-liberal regimes have been ruling in capitalist nations for years.
4. Hence, this distinction may seem as being too rudimentary to hold true in a complex contemporary society.

✖

Solution:
Correct Answer : 2341
Your Answer : 2134

Answer key/Solution

The correct sequence is 2341. Sentence 2 states that left supports socialism. "However" is used to shift the tone. In sentence 3, the author states that left has been ruling in capitalist nations. Hence 23 is a mandatory pair. Sentence 4 which starts with hence has to come after 2 and 3 as the distinction is made in sentence 2. Sentence 1 concludes this.

Bookmark FeedBack

Q.20 [11594329]

Directions for question (20): The passage given below is followed by four summaries. Choose the option that best captures the author's position.

The flow of plastic into the ocean is expected to triple by 2040 if current trends continue, up to 29m tonnes a year – the equivalent of 50kg for every metre of coastline in the world. All efforts made and announced so far to curb plastic waste are expected to reduce the volume by only 7%. Once in the sea, it breaks down into microplastics, to be ingested by marine life. Political momentum for a global agreement to address the full lifecycle of plastic has been growing, with several high-level ministerial declarations in favour.

1 ○ Once plastic enters the sea, it breaks down into microplastics that is ingested by marine life, causing irreparable damage to the ecosystem.

2 ○ Amid the continuing growth in the quantity of plastic waste discharged into the seas, there has been a movement towards a comprehensive, global response to the issue.

3 ○ Rather than dealing only with the disposal of plastic waste, governments worldwide now recognize the need to address the full life cycle of plastic.

4 ○ All efforts made till now to curb plastic waste have been in vain and the political establishment needs to take immediate cognizance of this fact.

✖

Answer key/Solution

Solution:
Correct Answer : 2
Your Answer : 1
The passage focuses on two aspects: the expected increase in the amount of plastic waste being discharged into the oceans across the world and how the governments across the world now recognize the need to address the full lifecycle of plastic. This makes option 2 correct. Option 1 addresses only a part of the passage. Option 3 distorts the intended meaning while option 4 cannot be ascertained from the information provided.

Bookmark FeedBack

Directions for question (21): The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

Q.21 [11594329]

1. Normally, it flies engineers to its clients' factories to identify and solve quality-control problems.
2. That's not an option in a world where flights are grounded, borders closed, and security tightened.
3. The UK company P2i adds water-repelling nanocoatings to smartphones and other gadgets.
4. So in some plants, P2i now relies on a system that uses artificial intelligence to look for even the slightest defects.

✖

Answer key/Solution

Solution:
Correct Answer : 3124
Your Answer : 3421
Statement 3 is the opening sentence of the paragraph. This is followed by statement 1 as 'it' in statement 1 refers to 'the UK company' in statement 3. Then statement 2 follows as it says that flying engineers to its clients is not possible as the flights are grounded, and borders are closed. Statement 4 talks about what the company does now. Thus, 3124 is the correct sequence.

Bookmark FeedBack

Five sentences related to a topic are given below. Four of them can be put together to form a meaningful and coherent short paragraph. Identify the odd one out.

Q.22 [11594329]

1. But when a cyberattack is actually launched by either the government or individual, there is often no real consequence because of the anonymity it offers.
2. Individuals are just as capable of launching cyberattacks, abetted and encouraged by their governments.
3. Malware is often simple, low-cost and difficult to trace.
4. As a large number of infected devices generates a huge amount of cryptocurrency, cybercriminal see this as a lucrative crime.
5. A lot of malware, for example, is manufactured in China, and there is reported collaboration between the People's Liberation Army (PLA) and Chinese universities to train hackers.

✖

Answer key/Solution

Solution:
Correct Answer : 4
Your Answer : 3
2351 is the correct sequence. Options (2) and (3) form a pair as option (2) talk about cyberattacks which is carried out through malware mentioned in option (3). Option (5) continues to talk about malware and uses China as an example of a country that manufactures malware. Option (1) follows option (5) because it also talks about cyberattack. However, option (4) talks about cryptocurrency. Thus, option (4) is the odd one out.

Bookmark FeedBack

Q.23 [11594329]

Directions for question (23): The passage given below is followed by four summaries. Choose the option that best captures the author's position.

Uncertainty, confusion, fear: we've known all these and more, in bucketloads, since March. But how have our children fared? What are the long-term effects of the Corona pandemic on them, and how can we help them navigate it all, whatever their age? According to the experts, kids are resilient, and while we won't know for a long time how affected they've been by the pandemic, they are wired to cope. However, children who have already experienced difficulties, especially mental health issues, are at particular risk. Just as important is your own mental health, which will have a huge effect on how your child deals with what's happening, so don't forget to look after your own needs, and to seek help – from friends, your doctor, a therapist – if you're finding it tough. Be centred on – and guided by – your child. Each one is different, and each child's reaction to the pandemic is different. Start with them, rather than with what's happening out there in the big world.

1 ○ The education system and its impact on children during the pandemic

2 ○ Insanity and the challenge of maintaining a mental equilibrium during the pandemic

3 ○ The world is going for a toss during the pandemic and children are faring the worst.

4 ○ The importance of focusing on your mental health and that of your child through the pandemic

✔

Answer key/Solution

Solution:
Correct Answer : 4
Your Answer : 4
The passage talks about parenting during the pandemic and the possible challenges. Hence, option (4) is the most appropriate. The other options do not have any thematic relevance to the passage.

Bookmark FeedBack

Directions for question (24): The four sentences (labelled 1, 2, 3, and 4) given in this question, when properly sequenced, form a coherent paragraph. Decide on the proper order for the sentences and key in this sequence of four numbers as your answer.

Q.24 [11594329]

1. New wave, early synth pop, the new romantics, London's Blitz kids, OMD, David Bowie, Gary Numan, Visage, Depeche Mode, Soft Cell and many other groups from the post-industrial early 80s ushered in the first generation of popular music that was entirely created with electronic instruments.
4. Initially niche, it soon went mainstream and travelled across the continent, resulting in a new wave of homegrown German electronic acts such as Boytronic, Rheingold, DAF and Stratis.
2. In 1988, Virgin released Techno! The New Dance Sound of Detroit, compiled by Neil Rushton.
3. It introduced Europe to the first wave of Detroit techno artists, Blake Baxter, Derrick May and Eddie Fowlkes among them.

✖

Answer key/Solution

Solution:
Correct Answer : 1423
Your Answer : 1234
The correct sequence is 1423. Sentence 4 has to follow sentence 1. "it" in 4 refers to the new wave. Sentence 4 talks about yet another new wave. Sentence 4 refers to a wave of popular music travelling across the continent and sentence 3 talks about Europe being introduced to the first wave of Detroit which is mentioned in 2. Hence, 23 a mandatory pair will follow 14. Thus, 1423 is the best sequence.

Bookmark FeedBack

Q.25 [11594329]

Directions for question (25): The passage given below is followed by four summaries. Choose the option that best captures the author's position.

The Assam government on Tuesday announced a slew of benefits for students, teachers and educational institutions in the state. The measures include provincialisation (taking over of liabilities like payment of salaries and other benefits by the government) of schools, junior colleges and colleges, teachers, professors, gifts of scooters to meritorious girl students, appointment of new teachers etc. "We had started provincialisation of educational institutions in 2011, but it nearly stopped in 2014. Today after a gap of 6 years we are releasing the first list of provincialised institutes as per notification issued by Governor Jagdish Mukhi," said health, finance, education and PWD minister Himanta Biswa Sarma in Guwahati. Initially, 197 high schools, 9 higher secondary schools, 9 junior colleges and 27 colleges would get provincialised. Jobs of 3049 teachers in high schools and 931 professors in colleges would also get provincialised, informed Sarma.

1 ○ The Assam government's initiatives to revamp pedagogy and school infrastructure

2 ○ The Assam government's promise of sops to students, teachers and institutions of learning

3 ○ The Assam government's policymaking related to education in the state

4 ○ The Assam government's perception of education and how education can be spread in the state.

✔

Answer key/Solution

Solution:
Correct Answer : 2
Your Answer : 2
The passage clearly talks about the measures taken by the Assam government that will benefit students, teachers and educational institutions. Hence, option (2) is correct. The measures don't include pedagogy or the methodology of teaching. Hence, option (1) is false. Options (3) and (4) are out of scope and hence, incorrect.

Bookmark FeedBack

Directions for question (26): Five sentences related to a topic are given below. Four of them can be put together to form a meaningful and coherent short paragraph. Identify the odd one out.

Q.26 [11594329]

1. The most important driver of the Quad's evolution is China.
2. Its military aggression against India and Japan, its attempts to subvert Australian civil society and its embrace of open geopolitical rivalry with the US is the strategic glue of the Quad.
3. Despite the US push to formalise the Quad, the grouping continues to be loose and informal in nature, which has raised questions among experts as to whether it can effectively work as a counterweight to China.
4. The most important determinant of which path is taken is the degree to which the Quad members can disentangle themselves from the economic and technological embrace of China.
5. Which is why discussions on 5G networks, supply chains and rare earths are so important to defining how far the Quad can go on the military and strategic side.

✖

Solution:

Answer key/Solution

Correct Answer : 3

Your Answer : 2

1245 is the correct sequence. Option (1) cites the factor that led to the formation of Quad and option (2) continues with that by providing additional information. Option (2) is further followed by option (4) as option (4) talks about the degree to which Quad members can disentangle themselves from the economic and technological embrace of China. Option (4) is followed by option (5) as it talks about 5G networks and supply chains which are directly related to technology and economy mentioned in option (4). In short, 1245 together look at the origin and strength of Quad. However, option (3) casts doubt on the capability of Quad. Thus, option (3) is the odd one out.

Bookmark FeedBack

Sec 2

Directions for questions 27 to 30: Answer the questions on the basis of the information given below.

Bakers Oven supplied cupcakes and doughnuts for three birthday parties, which were being celebrated at the houses of Mary, Nancy and Patsy. Bakers Oven supplies only two flavors - chocolate and vanilla - of cupcakes and doughnuts. The total number of cupcakes and doughnuts supplied to all three parties is 700.

It is also known that:
• Exactly 45% of the total order was for cupcakes.
• The total order for Patsy's party was 90 more than the total order for Mary's party.
• Nancy ordered an equal number of cupcakes and doughnuts and the number of vanilla cupcakes supplied to her was ten less than the number of vanilla doughnuts.
• Exactly 30% of the total order was for Mary. The number of chocolate doughnuts ordered by her was double the number of chocolate cupcakes ordered by her.
• Patsy ordered 50% of the total number of chocolate doughnuts.
The table gives details of few items supplied to the three birthday parties.

	Cupcakes	Chocolate flavored items
Mary	80	90
Nancy	-	110
Patsy	140	-

Q.27 [11594329]
Who ordered the maximum number of vanilla cupcakes?

1 ○ Mary

2 ○ Nancy

3 ○ Patsy

4 ○ Cannot be determined

✖

Solution:
Correct Answer : 4
Your Answer : 3

Answer key/Solution

According to the given information:
The total number of cupcakes and doughnuts supplied to all three parties is 700 and 45% of the total order was for cupcakes. Also 30% of the total order was for Mary. The total order for Patsy's party was 90 more than the total order for Mary's party.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary					30% of 700 = 210
Nancy					700 – 300 – 210 = 190
Patsy					210 + 90 = 300
Total	45% of 700 = 315		700 – 315 = 385		700

Nancy:
From the table we can see that out of 190 items ordered by Nancy, 110 were chocolate flavored items. So the number of vanilla flavored items was 80. Now it is given that the number of vanilla cupcakes supplied to her was ten less than the number of vanilla doughnuts. So the number of vanilla cupcakes and doughnuts is 35 and 45 respectively.
Nancy ordered an equal number of cupcakes and doughnuts, which is 1/2 of 190 = 95.
So the number of chocolate cupcakes and chocolate doughnuts ordered by Nancy will be 60 and 50 respectively.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary					210
Nancy	60	35	50	45	190
Patsy					300
Total	315		385		700

Mary:
We see that the number of chocolate flavored items ordered by Mary is 90. Also the number of chocolate doughnuts ordered by Mary was double the number of chocolate cupcakes ordered by her. So Mary orders 60 chocolate doughnuts and 30 chocolate cupcakes. From the table it is clear that Mary orders a total of 80 cupcakes so number of chocolate cupcakes will be 50.
Hence, the number of vanilla doughnuts for Mary will be = 210 – (30 + 50 + 60) = 70.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary	30	50	60	70	210
Nancy	60	35	50	45	190
Patsy					300
Total	315		385		700

Patsy:
Patsy ordered 50% of the total number of chocolate doughnuts, which will be 110. Also the number of vanilla doughnuts ordered by Patsy can be found by 385 – (60 + 70 + 50 + 45 + 110) = 50.

Total number of cupcakes ordered by Patsy is 140.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary	30	50	60	70	210
Nancy	60	35	50	45	190
Patsy	140		110	50	300
Total	315		385		700

We cannot uniquely determine the number of vanilla cupcakes ordered by Patsy.

Bookmark FeedBack

Directions for questions 27 to 30: Answer the questions on the basis of the information given below.

Bakers Oven supplied cupcakes and doughnuts for three birthday parties, which were being celebrated at the houses of Mary, Nancy and Patsy. Bakers Oven supplies only two flavors - chocolate and vanilla - of cupcakes and doughnuts. The total number of cupcakes and doughnuts supplied to all three parties is 700.

It is also known that:
• Exactly 45% of the total order was for cupcakes.
• The total order for Patsy's party was 90 more than the total order for Mary's party.
• Nancy ordered an equal number of cupcakes and doughnuts and the number of vanilla cupcakes supplied to her was ten less than the number of vanilla doughnuts.
• Exactly 30% of the total order was for Mary. The number of chocolate doughnuts ordered by her was double the number of chocolate cupcakes ordered by her.
• Patsy ordered 50% of the total number of chocolate doughnuts.
The table gives details of few items supplied to the three birthday parties.

	Cupcakes	Chocolate flavored items
Mary	80	90
Nancy	-	110
Patsy	140	-

Q.28 [11594329]
What was the total number chocolate flavored doughnuts ordered by all three of them?

1 ○ 165

2 ○ 220

3 ○ 170

4 ○ 190

✖

Solution:
Correct Answer : 2
Your Answer : 4

Answer key/Solution

According to the given information:
The total number of cupcakes and doughnuts supplied to all three parties is 700 and 45% of the total order was for cupcakes. Also 30% of the total order was for Mary. The total order for Patsy's party was 90 more than the total order for Mary's party.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary					30% of 700 = 210
Nancy					700 – 300 – 210 = 190
Patsy					210 + 90 = 300
Total	45% of 700 = 315		700 – 315 = 385		700

Nancy:
From the table we can see that out of 190 items ordered by Nancy, 110 were chocolate flavored items. So the number of vanilla flavored items was 80. Now it is given that the number of vanilla cupcakes supplied to her was ten less than the number of vanilla doughnuts. So the number of vanilla cupcakes and doughnuts is 35 and 45 respectively.
Nancy ordered an equal number of cupcakes and doughnuts, which is 1/2 of 190 = 95.
So the number of chocolate cupcakes and chocolate doughnuts ordered by Nancy will be 60 and 50 respectively.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary					210
Nancy	60	35	50	45	190
Patsy					300
Total	315		385		700

Mary:
We see that the number of chocolate flavored items ordered by Mary is 90. Also the number of chocolate doughnuts ordered by Mary was double the number of chocolate cupcakes ordered by her. So Mary orders 60 chocolate doughnuts and 30 chocolate cupcakes. From the table it is clear that Mary orders a total of 80 cupcakes so number of chocolate cupcakes will be 50.
Hence, the number of vanilla doughnuts for Mary will be = 210 – (30 + 50 + 60) = 70.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary	30	50	60	70	210
Nancy	60	35	50	45	190
Patsy					300
Total	315		385		700

Patsy:
Patsy ordered 50% of the total number of chocolate doughnuts, which will be 110. Also the number of vanilla doughnuts ordered by Patsy can be found by 385 – (60 + 70 + 50 + 45 + 110) = 50.

Total number of cupcakes ordered by Patsy is 140.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary	30	50	60	70	210
Nancy	60	35	50	45	190
Patsy	140		110	50	300
Total	315		385		700

The total number of chocolate flavored doughnuts ordered by all three of them is 60 + 50 + 110 = 220.

Bookmark FeedBack

Directions for questions 27 to 30: Answer the questions on the basis of the information given below.

Bakers Oven supplied cupcakes and doughnuts for three birthday parties, which were being celebrated at the houses of Mary, Nancy and Patsy. Bakers Oven supplies only two flavors - chocolate and vanilla - of cupcakes and doughnuts. The total number of cupcakes and doughnuts supplied to all three parties is 700.

It is also known that:
• Exactly 45% of the total order was for cupcakes.
• The total order for Patsy's party was 90 more than the total order for Mary's party.
• Nancy ordered an equal number of cupcakes and doughnuts and the number of vanilla cupcakes supplied to her was ten less than the number of vanilla doughnuts.
• Exactly 30% of the total order was for Mary. The number of chocolate doughnuts ordered by her was double the number of chocolate cupcakes ordered by her.
• Patsy ordered 50% of the total number of chocolate doughnuts.
The table gives details of few items supplied to the three birthday parties.

	Cupcakes	Chocolate flavored items
Mary	80	90
Nancy	-	110
Patsy	140	-

Q.29 [11594329]
If 20% of the total order at Patsy's house was for vanilla cupcakes, then what was the difference in the total number of chocolate doughnuts and chocolate cupcakes supplied by Baker's Oven to these three parties?

1 ○ 50

2 ○ 20

3 ○ 45

4 ○ 65

✖

Solution:
Correct Answer : 1
Your Answer : 2

Answer key/Solution

According to the given information:
The total number of cupcakes and doughnuts supplied to all three parties is 700 and 45% of the total order was for cupcakes.
Also 30% of the total order was for Mary. The total order for Patsy's party was 90 more than the total order for Mary's party.

	Cupcakes	Doughnuts	

	Chocolate	Vanilla	Chocolate	Vanilla	Total
Mary					30% of 700 = 210
Nancy					700 – 300 – 210 = 190
Patsy					210 + 90 = 300
Total	45% of 700 = 315		700 – 315 = 385		700

Nancy:
From the table we can see that out of 190 items ordered by Nancy, 110 were chocolate flavored items. So the number of vanilla flavored items was 80. Now it is given that the number of vanilla cupcakes supplied to her was ten less than the number of vanilla doughnuts. So the number of vanilla cupcakes and doughnuts is 35 and 45 respectively.
Nancy ordered an equal number of cupcakes and doughnuts, which is 1/2 of 190 = 95.
So the number of chocolate cupcakes and chocolate doughnuts ordered by Nancy will be 60 and 50 respectively.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary					210
Nancy	60	35	50	45	190
Patsy					300
Total	315		385		700

Mary:
We see that the number of chocolate flavored items ordered by Mary is 90. Also the number of chocolate doughnuts ordered by Mary was double the number of chocolate cupcakes ordered by her. So Mary orders 60 chocolate doughnuts and 30 chocolate cupcakes. From the table it is clear that Mary orders a total of 80 cupcakes so number of chocolate cupcakes will be 50.
Hence, the number of vanilla doughnuts for Mary will be = 210 – (30 + 50 + 60) = 70.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary	30	50	60	70	210
Nancy	60	35	50	45	190
Patsy					300
Total	315		385		700

Patsy:
Patsy ordered 50% of the total number of chocolate doughnuts, which will be 110. Also the number of vanilla doughnuts ordered by Patsy can be found by 385 – (60 + 70 + 50 + 45 + 110) = 50.

Total number of cupcakes ordered by Patsy is 140.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary	30	50	60	70	210
Nancy	60	35	50	45	190
Patsy	140		110	50	300
Total	315		385		700

Given that 20% of the total order at Patsy's house was for vanilla cupcakes.
⇒ Number of vanilla cupcakes = 20% of 300 = 60 ⇒ Chocolate cupcakes = 140 – 60 = 80.
∴ Total number of chocolate doughnuts and chocolate cupcakes supplied by Baker's Oven is 220 and 170 respectively.
Hence, required difference = 220 – 170 = 50.

Bookmark FeedBack

Directions for questions 27 to 30: Answer the questions on the basis of the information given below.

Bakers Oven supplied cupcakes and doughnuts for three birthday parties, which were being celebrated at the houses of Mary, Nancy and Patsy. Bakers Oven supplies only two flavors - chocolate and vanilla - of cupcakes and doughnuts. The total number of cupcakes and doughnuts supplied to all three parties is 700.

It is also known that:
• Exactly 45% of the total order was for cupcakes.
• The total order for Patsy's party was 90 more than the total order for Mary's party.
• Nancy ordered an equal number of cupcakes and doughnuts and the number of vanilla cupcakes supplied to her was ten less than the number of vanilla doughnuts.
• Exactly 30% of the total order was for Mary. The number of chocolate doughnuts ordered by her was double the number of chocolate cupcakes ordered by her.
• Patsy ordered 50% of the total number of chocolate doughnuts.
The table gives details of few items supplied to the three birthday parties.

	Cupcakes	Chocolate flavored items
Mary	80	90
Nancy	-	110
Patsy	140	-

Q.30 [11594329]
What was the total number of vanilla doughnuts supplied by Baker's Oven to these three parties?

1 ○ 140

2 ○ 220

3 ○ 165

4 ○ 145

Solution:
Correct Answer : 3

Answer key/Solution

According to the given information:
The total number of cupcakes and doughnuts supplied to all three parties is 700 and 45% of the total order was for cupcakes. Also 30% of the total order was for Mary. The total order for Patsy's party was 90 more than the total order for Mary's party.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary					30% of 700 = 210
Nancy					700 – 300 – 210 = 190
Patsy					210 + 90 = 300
Total	45% of 700 = 315		700 – 315 = 385		700

Nancy:
From the table we can see that out of 190 items ordered by Nancy, 110 were chocolate flavored items. So the number of vanilla flavored items was 80. Now it is given that the number of vanilla cupcakes supplied to her was ten less than the number of vanilla doughnuts. So the number of vanilla cupcakes and doughnuts is 35 and 45 respectively.
Nancy ordered an equal number of cupcakes and doughnuts, which is 1/2 of 190 = 95.
So the number of chocolate cupcakes and chocolate doughnuts ordered by Nancy will be 60 and 50 respectively.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary					210
Nancy	60	35	50	45	190
Patsy					300
Total	315		385		700

Mary:
We see that the number of chocolate flavored items ordered by Mary is 90. Also the number of chocolate doughnuts ordered by Mary was double the number of chocolate cupcakes ordered by her. So Mary orders 60 chocolate doughnuts and 30 chocolate cupcakes. From the table it is clear that Mary orders a total of 80 cupcakes so number of chocolate cupcakes will be 50.
Hence, the number of vanilla doughnuts for Mary will be = 210 – (30 + 50 + 60) = 70.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary	30	50	60	70	210
Nancy	60	35	50	45	190
Patsy					300
Total	315		385		700

Patsy:
Patsy ordered 50% of the total number of chocolate doughnuts, which will be 110. Also the number of vanilla doughnuts ordered by Patsy can be found by 385 – (60 + 70 + 50 + 45 + 110) = 50.

Total number of cupcakes ordered by Patsy is 140.

	Cupcakes		Doughnuts		Total
	Chocolate	Vanilla	Chocolate	Vanilla	
Mary	30	50	60	70	210
Nancy	60	35	50	45	190
Patsy	140		110	50	300
Total	315		385		700

Total number of vanilla doughnuts supplied by Baker's Oven = 70 + 45 + 50 = 165.

Bookmark FeedBack

Directions for questions 31 to 34: Answer the questions on the basis of the information given below.

Each of the five friends – Adi, Brendon, Charlie, Dany, and Eoin studies in a different college among KMC, RSD, MSC, JMC, and BVP and lives in a different locality among KC, RG, NC, SDA, and GK. Each person has a different bike among Splendor, Freedom, Pulsar, Karizma, and Rxz and each plays a different sport among Cricket, Tennis, Football, Basketball, and Rugby.

Further, it is known that:
(i) The person who lives in RG does not study at KMC college and the person who has Splendor does not play Rugby.
(ii) Dany plays Rugby and Charlie who lives in SDA, studies at MSC college.
(iii) Adi, who plays Basketball, does not has Karizma and the person who lives in KC plays Tennis. Further, the person who lives in GK does not play Cricket.
(iv)The person who plays Cricket, does not study at MSC college and he has Pulsar bike.
(v) The person who lives in NC, studies at RSD college and has Karizma bike.
(vi) Brendon, who has Rxz, does not play Football and the person who has Freedom, studies at BVP college.

Q.31 [11594329]
Who lives in RG?

1 ○ Adi

2 ○ Brendon

3 ○ Dany

4 ○ Eoin

✔

Solution:
Correct Answer : 4
Your Answer : 4

Answer key/Solution

According to the given information:
From statement (ii), Dany plays Rugby; from statement (iii), Adi plays Basketball; from statement (iv), the person with Pulsar plays Cricket.
From statement (vi), Brendon has Rxz and he does not play Football. Hence, it can be concluded that Brendon plays Tennis.
From statement (iii), it can be concluded that Brendon lives in KC.
From statements (ii) and (iv), it can be concluded that Charlie, who lives in SDA and studies at MSC college, does not play Cricket and the only option of sport left for him is Football.
From statements (iv), (v), and (vi), it can be concluded that Charlie does not has Pulsar, Karizma, Rxz, and Freedom, hence, the only option of bike left for him is Splendor.
From statement (iii), Adi does not has Karizma, hence, the only bike left for him is Freedom and therefore, it can be concluded that Dany has Karizma bike.
From statement (vi), it can be further concluded that Adi studies at BVP college. And from statement (v), it can be concluded that Dany lives in NC and studies at RSD college.
So the following table can be made:

Friend	College	Locality	Bike	Sport
Adi	BVP		× Karizma; × Pulsor; Freedom	Basketball
Brendon		KC	Rxz	Tennis; × Football
Charlie	MSC	SDA	× Pulsor; × Feedom; × Karizma; Splendor	Football; × Cricket
Dany	RSD	NC	× Spendor; × Pulsor; Karizma	Rugby
Eoin			Pulsar	Cricket

From Statement (iii), it can be concluded that Eoin does not live in GK, hence, the only option left for him is RG and therefore, the person living in GK has to be Adi.

And from statement (i), it can be concluded that Eoin studies at JMC and therefore, it has to be Brendon, who studies at KMC.
Hence, the final table is as following:

Friend	College	Locality	Bike	Sport
Adi	BVP	GK	Freedom	Basketball
Brendon	KMC	KC	Rxz	Tennis
Charlie	MSC	SDA	Splendor	Football
Dany	RSD	NC	Karizma	Rugby
Eoin	JMC	RG	Pulsar	Cricket

Eoin lives in RG.

Bookmark FeedBack

Directions for questions 31 to 34: Answer the questions on the basis of the information given below.

Each of the five friends – Adi, Brendon, Charlie, Dany, and Eoin studies in a different college among KMC, RSD, MSC, JMC, and BVP and lives in a different locality among KC, RG, NC, SDA, and GK. Each person has a different bike among Splendor, Freedom, Pulsar, Karizma, and Rxz and each plays a different sport among Cricket, Tennis, Football, Basketball, and Rugby.

Further, it is known that:
(i) The person who lives in RG does not study at KMC college and the person who has Splendor does not play Rugby.
(ii) Dany plays Rugby and Charlie who lives in SDA, studies at MSC college.
(iii) Adi, who plays Basketball, does not has Karizma and the person who lives in KC plays Tennis. Further, the person who lives in GK does not play Cricket.
(iv)The person who plays Cricket, does not study at MSC college and he has Pulsar bike.
(v) The person who lives in NC, studies at RSD college and has Karizma bike.
(vi) Brendon, who has Rxz, does not play Football and the person who has Freedom, studies at BVP college.

Q.32 [11594329]
Which sport does the person who lives in GK play?

1 ○ Tennis

2 ○ Football

3 ○ Rugby

4 ○ Basketball

✔

Solution:
Correct Answer : 4
Your Answer : 4

Answer key/Solution

According to the given information:
From statement (ii), Dany plays Rugby; from statement (iii), Adi plays Basketball; from statement (iv), the person with Pulsar plays Cricket.
From statement (vi), Brendon has Rxz and he does not play Football. Hence, it can be concluded that Brendon plays Tennis.
From statement (iii), it can be concluded that Brendon lives in KC.
From statements (ii) and (iv), it can be concluded that Charlie, who lives in SDA and studies at MSC college, does not play Cricket and the only option of sport left for him is Football.
From statements (iv), (v), and (vi), it can be concluded that Charlie does not has Pulsar, Karizma, Rxz, and Freedom, hence, the only option of bike left for him is Splendor.
From statement (iii), Adi does not has Karizma, hence, the only bike left for him is Freedom and therefore, it can be concluded that Dany has Karizma bike.
From statement (vi), it can be further concluded that Adi studies at BVP college. And from statement (v), it can be concluded that Dany lives in NC and studies at RSD college.
So the following table can be made:

Friend	College	Locality	Bike	Sport
Adi	BVP		× Karizma; × Pulsor; Freedom	Basketball
Brendon		KC	Rxz	Tennis; × Football
Charlie	MSC	SDA	× Pulsor; × Feedom; × Karizma; Splendor	Football; × Cricket
Dany	RSD	NC	× Spendor; × Pulsor; Karizma	Rugby
Eoin			Pulsar	Cricket

From Statement (iii), it can be concluded that Eoin does not live in GK, hence, the only option left for him is RG and therefore, the person living in GK has to be Adi.

And from statement (i), it can be concluded that Eoin studies at JMC and therefore, it has to be Brendon, who studies at KMC.
Hence, the final table is as following:

Friend	College	Locality	Bike	Sport
Adi	BVP	GK	Freedom	Basketball
Brendon	KMC	KC	Rxz	Tennis
Charlie	MSC	SDA	Splendor	Football
Dany	RSD	NC	Karizma	Rugby
Eoin	JMC	RG	Pulsar	Cricket

The person who lives in GK plays Basketball.

Bookmark FeedBack

Directions for questions 31 to 34: Answer the questions on the basis of the information given below.

Each of the five friends – Adi, Brendon, Charlie, Dany, and Eoin studies in a different college among KMC, RSD, MSC, JMC, and BVP and lives in a different locality among KC, RG, NC, SDA, and GK. Each person has a different bike among Splendor, Freedom, Pulsar, Karizma, and Rxz and each plays a different sport among Cricket, Tennis, Football, Basketball, and Rugby.

Further, it is known that:
(i) The person who lives in RG does not study at KMC college and the person who has Splendor does not play Rugby.
(ii) Dany plays Rugby and Charlie who lives in SDA, studies at MSC college.
(iii) Adi, who plays Basketball, does not has Karizma and the person who lives in KC plays Tennis. Further, the person who lives in GK does not play Cricket.
(iv)The person who plays Cricket, does not study at MSC college and he has Pulsar bike.
(v) The person who lives in NC, studies at RSD college and has Karizma bike.
(vi) Brendon, who has Rxz, does not play Football and the person who has Freedom, studies at BVP college.

Q.33 [11594329]
The person who has Splendor bike lives in which locality?

1 ○ SDA

2 ○ GK

3 ○ KC

4 ○ RG

✔

Solution:
Correct Answer : 1
Your Answer : 1

Answer key/Solution

According to the given information:
From statement (ii), Dany plays Rugby; from statement (iii), Adi plays Basketball; from statement (iv), the person with Pulsar plays Cricket.
From statement (vi), Brendon has Rxz and he does not play Football. Hence, it can be concluded that Brendon plays Tennis.
From statement (iii), it can be concluded that Brendon lives in KC.
From statements (ii) and (iv), it can be concluded that Charlie, who lives in SDA and studies at MSC college, does not play Cricket and the only option of sport left for him is Football.
From statements (iv), (v), and (vi), it can be concluded that Charlie does not has Pulsar, Karizma, Rxz, and Freedom, hence, the only option of bike left for him is Splendor.
From statement (iii), Adi does not has Karizma, hence, the only bike left for him is Freedom and therefore, it can be concluded that Dany has Karizma bike.
From statement (vi), it can be further concluded that Adi studies at BVP college. And from statement (v), it can be concluded that Dany lives in NC and studies at RSD college.
So the following table can be made:

Friend	College	Locality	Bike	Sport
Adi	BVP		× Karizma; × Pulsor; Freedom	Basketball
Brendon		KC	Rxz	Tennis; × Football
Charlie	MSC	SDA	× Pulsor; × Feedom; × Karizma; Splendor	Football; × Cricket
Dany	RSD	NC	× Spendor; × Pulsor; Karizma	Rugby
Eoin			Pulsar	Cricket

From Statement (iii), it can be concluded that Eoin does not live in GK, hence, the only option left for him is RG and therefore, the person living in GK has to be Adi.

And from statement (i), it can be concluded that Eoin studies at JMC and therefore, it has to be Brendon, who studies at KMC.
Hence, the final table is as following:

Friend	College	Locality	Bike	Sport
Adi	BVP	GK	Freedom	Basketball
Brendon	KMC	KC	Rxz	Tennis
Charlie	MSC	SDA	Splendor	Football
Dany	RSD	NC	Karizma	Rugby
Eoin	JMC	RG	Pulsar	Cricket

The person who has Splendor bike lives in SDA locality.

Bookmark FeedBack

Directions for questions 31 to 34: Answer the questions on the basis of the information given below.

Each of the five friends – Adi, Brendon, Charlie, Dany, and Eoin studies in a different college among KMC, RSD, MSC, JMC, and BVP and lives in a different locality among KC, RG, NC, SDA, and GK. Each person has a different bike among Splendor, Freedom, Pulsar, Karizma, and Rxz and each plays a different sport among Cricket, Tennis, Football, Basketball, and Rugby.

Further, it is known that:
(i) The person who lives in RG does not study at KMC college and the person who has Splendor does not play Rugby.
(ii) Dany plays Rugby and Charlie who lives in SDA, studies at MSC college.
(iii) Adi, who plays Basketball, does not has Karizma and the person who lives in KC plays Tennis. Further, the person who lives in GK does not play Cricket.
(iv)The person who plays Cricket, does not study at MSC college and he has Pulsar bike.
(v) The person who lives in NC, studies at RSD college and has Karizma bike.
(vi) Brendon, who has Rxz, does not play Football and the person who has Freedom, studies at BVP college.

Q.34 [11594329]
Who studies at KMC college?

1 ○ Adi

2 ○ Brendon

3 ○ Dany

4 ○ Eoin

✔

Solution:
Correct Answer : 2
Your Answer : 2

Answer key/Solution

According to the given information:
From statement (ii), Dany plays Rugby; from statement (iii), Adi plays Basketball; from statement (iv), the person with Pulsar plays Cricket.
From statement (vi), Brendon has Rxz and he does not play Football. Hence, it can be concluded that Brendon plays Tennis.
From statement (iii), it can be concluded that Brendon lives in KC.
From statements (ii) and (iv), it can be concluded that Charlie, who lives in SDA and studies at MSC college, does not play Cricket and the only option of sport left for him is Football.
From statements (iv), (v), and (vi), it can be concluded that Charlie does not has Pulsar, Karizma, Rxz, and Freedom, hence, the only option of bike left for him is Splendor.
From statement (iii), Adi does not has Karizma, hence, the only bike left for him is Freedom and therefore, it can be concluded that Dany has Karizma bike.
From statement (vi), it can be further concluded that Adi studies at BVP college. And from statement (v), it can be concluded that Dany lives in NC and studies at RSD college.
So the following table can be made:

Friend	College	Locality	Bike	Sport
Adi	BVP		× Karizma; × Pulsor; Freedom	Basketball
Brendon		KC	Rxz	Tennis; × Football
Charlie	MSC	SDA	× Pulsor; × Feedom; × Karizma; Splendor	Football; × Cricket
Dany	RSD	NC	× Spendor; × Pulsor; Karizma	Rugby
Eoin			Pulsar	Cricket

From Statement (iii), it can be concluded that Eoin does not live in GK, hence, the only option left for him is RG and therefore, the person living in GK has to be Adi.

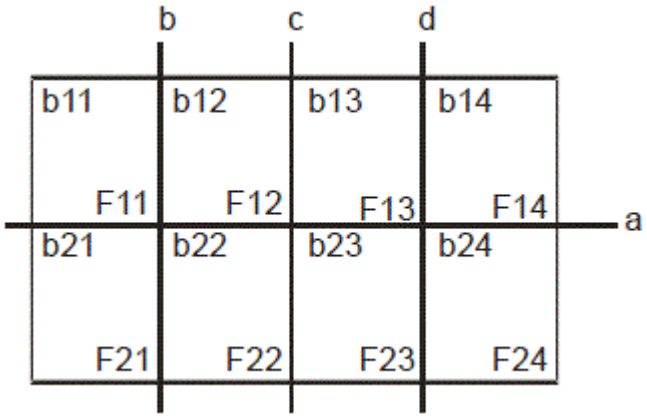
And from statement (i), it can be concluded that Eoin studies at JMC and therefore, it has to be Brendon, who studies at KMC.
Hence, the final table is as following:

Friend	College	Locality	Bike	Sport
Adi	BVP	GK	Freedom	Basketball
Brendon	KMC	KC	Rxz	Tennis
Charlie	MSC	SDA	Splendor	Football
Dany	RSD	NC	Karizma	Rugby
Eoin	JMC	RG	Pulsar	Cricket

Brendon studies at KMC college.

Bookmark FeedBack

Directions for questions 35 to 38: Answer the questions on the basis of the information given below.



A rectangular sheet is divided into a 4 × 2 grid of smaller rectangles on both sides, as shown in the figure. The numbers at the right hand bottom corner is the number written on the front side of each rectangle. The numbers at the left hand top corner is the number written on the back side of each rectangle. Three folds are made - the first about line a, then next about line c and the third about b.

The folded sheet is stitched to make a 16-page note book by slicing the edges using a knife. The sequence used for folding was:

1. Fold (a, top, bottom) : Fold about line a. Place top part on the bottom part of the fold.
2. Fold (c, right, left) : Fold about line c. Place the right part of the fold on the left part.
3. Fold (d, right, left) : Fold about line b. Place the right part of the fold on the left part.

(If 2 lines overlap, the one on the top is represented in the operation.)

The final note book had a sequence of pages from 1 to 16, 1 being the top most page and 16 the bottom most.

Q.35 [11594329]

The process of folding was followed as per the design. What is the page number of F22?

1 ○ 1

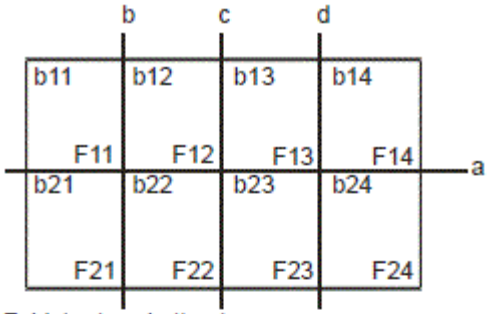
2 ○ 2

3 ○ 15

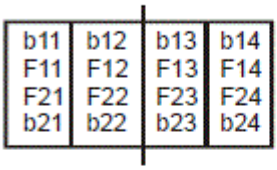
4 ○ 16

Solution:
Correct Answer : 2

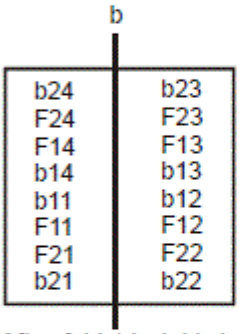
Answer key/Solution



Fold (a, top, bottom)



After fold (c, right, left)



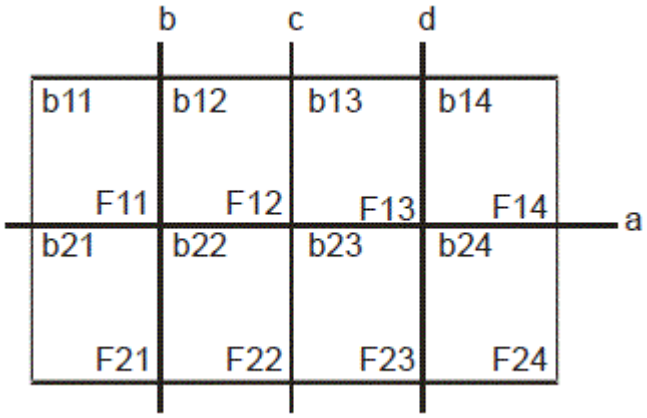
After fold (d, right, left)

1	b22
2	F22
3	F12
4	b12
5	b13
6	F13
7	F23
8	b23
9	b24
10	F24
11	F14
12	b14
13	b11
14	F11
15	F21
16	b21

The page number of F22 is 2.

Bookmark FeedBack

Directions for questions 35 to 38: Answer the questions on the basis of the information given below.



A rectangular sheet is divided into a 4 × 2 grid of smaller rectangles on both sides, as shown in the figure. The numbers at the right hand bottom corner is the number written on the front side of each rectangle. The numbers at the left hand top corner is the number written on the back side of each rectangle. Three folds are made - the first about line a, then next about line c and the third about b.

The folded sheet is stitched to make a 16-page note book by slicing the edges using a knife. The sequence used for folding was:

1. Fold (a, top, bottom) : Fold about line a. Place top part on the bottom part of the fold.
2. Fold (c, right, left) : Fold about line c. Place the right part of the fold on the left part.
3. Fold (d, right, left) : Fold about line b. Place the right part of the fold on the left part.

(If 2 lines overlap, the one on the top is represented in the operation.)

The final note book had a sequence of pages from 1 to 16, 1 being the top most page and 16 the bottom most.

Q.36 [11594329]

Two page numbers in the original sheet were misprinted, because of which the page numbers 3 and 12 got interchanged. Which 2 numbers were wrong in the original sheet?

1 ○ F12, F11

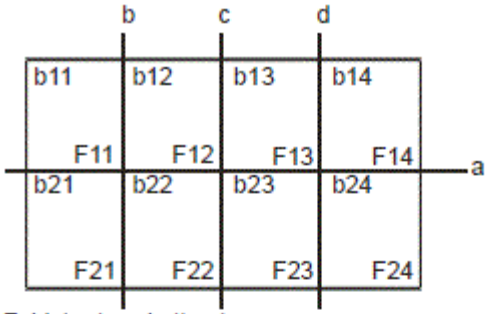
2 ○ F12, b14

3 ○ F23, b14

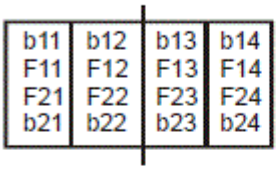
4 ○ F12, b11

Solution:
Correct Answer : 2

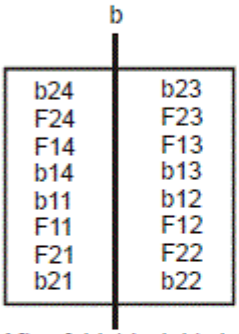
Answer key/Solution



Fold (a, top, bottom)



After fold (c, right, left)



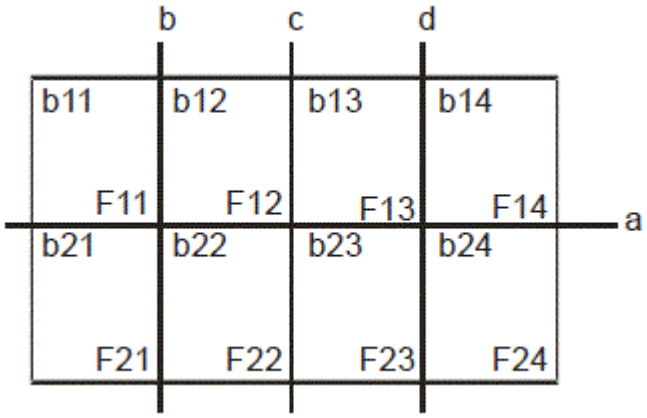
After fold (d, right, left)

1	b22
2	F22
3	F12
4	b12
5	b13
6	F13
7	F23
8	b23
9	b24
10	F24
11	F14
12	b14
13	b11
14	F11
15	F21
16	b21

F12, b14 numbers were wrong in the original sheet.

Bookmark FeedBack

Directions for questions 35 to 38: Answer the questions on the basis of the information given below.



A rectangular sheet is divided into a 4 × 2 grid of smaller rectangles on both sides, as shown in the figure. The numbers at the right hand bottom corner is the number written on the front side of each rectangle. The numbers at the left hand top corner is the number written on the back side of each rectangle. Three folds are made - the first about line a, then next about line c and the third about b.

The folded sheet is stitched to make a 16-page note book by slicing the edges using a knife. The sequence used for folding was:

1. Fold (a, top, bottom) : Fold about line a. Place top part on the bottom part of the fold.
2. Fold (c, right, left) : Fold about line c. Place the right part of the fold on the left part.
3. Fold (d, right, left) : Fold about line b. Place the right part of the fold on the left part.

(If 2 lines overlap, the one on the top is represented in the operation.)

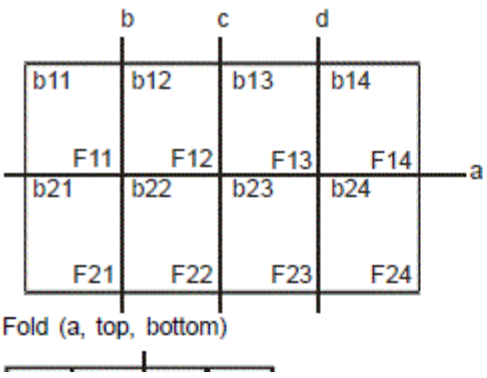
The final note book had a sequence of pages from 1 to 16, 1 being the top most page and 16 the bottom most.

Q.37 [11594329]

Additional information for questions 37 and 38: Ramu, who is new to the job, made a mistake in the sequence of folds. Hence, the page numbers got jumbled up. Ramu followed the following sequence while folding:
(a, top, bottom) → Fold (c, left, right) → Fold (b, left, right).
So all the page numbers had to be edited.
How many of the original page numbers had to be replaced by smaller page numbers?

Solution:
Correct Answer : 8

Answer key/Solution



b11	b12	b13	b14
F11	F12	F13	F14
F21	F22	F23	F24
b21	b22	b23	b24

After fold (c, right, left)

b

b24	b23
F24	F23
F14	F13
b14	b13
b11	b12
F11	F12
F21	F22
b21	b22

After fold (d, right, left)

1	b22
2	F22
3	F12
4	b12
5	b13
6	F13
7	F23
8	b23
9	b24
10	F24
11	F14
12	b14
13	b11
14	F11
15	F21
16	b21

Fold (c, left, right)

b22	b21
F22	F21
F12	F11
b12	b11
b13	b14
F13	F14
F23	F24
b23	b24

Fold (b, left, right)

1	b23
2	F23
3	F13
4	b13
5	b12
6	F12
7	F22
8	b22
9	b21

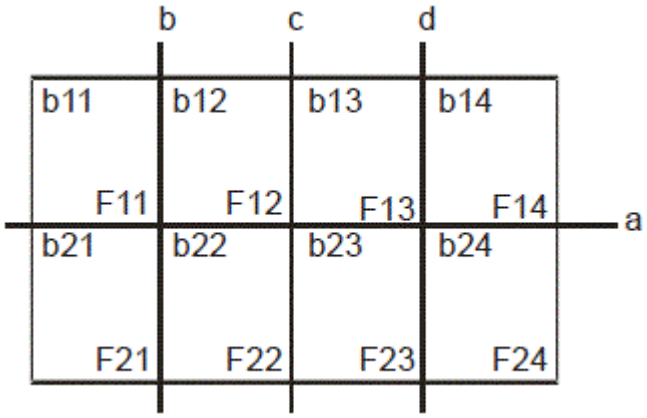
10	F21
11	F11
12	b11
13	b14
14	F14
15	F24
16	b24

b23, F23, F13, b13, b21, F21, F11, b11 were the page numbers that were edited into smaller numbers.

Changed page numbers	
b23	From 8 to 1
F23	From 7 to 2
F13	From 6 to 3
b13	From 5 to 4
b21	From 16 to 9
F21	From 15 to 10
F11	From 14 to 11
b11	From 13 to 12

Bookmark FeedBack

Directions for questions 35 to 38: Answer the questions on the basis of the information given below.



A rectangular sheet is divided into a 4 × 2 grid of smaller rectangles on both sides, as shown in the figure. The numbers at the right hand bottom corner is the number written on the front side of each rectangle. The numbers at the left hand top corner is the number written on the back side of each rectangle. Three folds are made - the first about line a, then next about line c and the third about b.

The folded sheet is stitched to make a 16-page note book by slicing the edges using a knife. The sequence used for folding was:

1. Fold (a, top, bottom) : Fold about line a. Place top part on the bottom part of the fold.
2. Fold (c, right, left) : Fold about line c. Place the right part of the fold on the left part.
3. Fold (d, right, left) : Fold about line b. Place the right part of the fold on the left part.

(If 2 lines overlap, the one on the top is represented in the operation.)

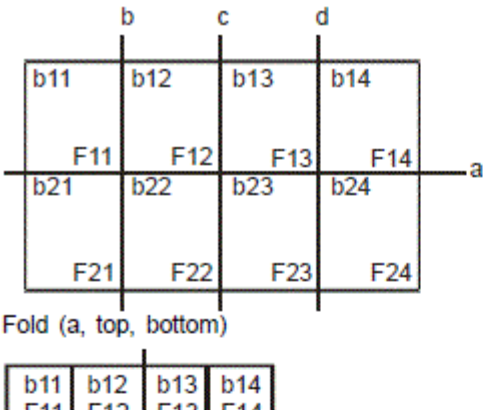
The final note book had a sequence of pages from 1 to 16, 1 being the top most page and 16 the bottom most.

Q.38 [11594329]

Additional information for questions 37 and 38: Ramu, who is new to the job, made a mistake in the sequence of folds. Hence, the page numbers got jumbled up. Ramu followed the following sequence while folding:
(a, top, bottom) → Fold (c, left, right) → Fold (b, left, right).
So all the page numbers had to be edited.
How many pairs of the original numbers, interchange their page numbers?

Solution:
Correct Answer : 8

Answer key/Solution



F11	F12	F13	F14
F21	F22	F23	F24
b21	b22	b23	b24

After fold (c, right, left)

b

b24	b23
F24	F23
F14	F13
b14	b13
b11	b12
F11	F12
F21	F22
b21	b22

After fold (d, right, left)

1	b22
2	F22
3	F12
4	b12
5	b13
6	F13
7	F23
8	b23
9	b24
10	F24
11	F14
12	b14
13	b11
14	F11
15	F21
16	b21

Fold (c, left, right)

b22	b21
F22	F21
F12	F11
b12	b11
b13	b14
F13	F14
F23	F24
b23	b24

Fold (b, left, right)

1	b23
2	F23
3	F13
4	b13
5	b12
6	F12
7	F22
8	b22
9	b21
10	F21

11	F11
12	b11
13	b14
14	F14
15	F24
16	b24

(b22, b23); (F22, F23); (F12, F13); (b12, b13); (b24, b21); (F24, F21); (F14, F11); (b14, b11).

Bookmark FeedBack

Directions for questions 39 to 42: Answer the questions on the basis of the information given below.

Rakesh was a keen stock investor. During the Covid lockdown, since there was more time at hand he invested in 4 stocks A, B, C, D during the month of May.
The following data is the 5 day moving average (5 DMA) of these 5 stocks. 5 day moving average is the average prices of stocks on the last 5 days (including that day) when the stocks were traded in the stock exchange.
The following table gives the 5 DMA of stocks (in Rs.) at 1200 hours on 8 consecutive days when the markets were open.

Stock	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8
A	540	543	543	542	541	540	542	543
B	1020	1025	1024	1025	1024	1026	1028	1029
C	141	143	144	143	142	143	142	141
D	232	233	235	236	237	233	237	236
E	800	804	806	812	815	813	816	810

Assume all transactions happen at 1200 hours, and ignore all other costs involved in the transactions.

Q.39 [11594329]
If Rakesh bought 1000 shares of D on day 2 and sold on day 7, what is his profit (in Rs.)?

1 ○ 800

2 ○ 4,000

3 ○ 8,000

4 ○ 20,000

Solution:

Correct Answer : 4

Answer key/Solution

D2 + D3 + D4 + D5 + D6 = 5 × 233
D3 + D4 + D5 + D6 + D7 = 5 × 237
D7 – D2 = Rs. 20
So, profit = Rs. 20,000.

Bookmark FeedBack

Directions for questions 39 to 42: Answer the questions on the basis of the information given below.

Rakesh was a keen stock investor. During the Covid lockdown, since there was more time at hand he invested in 4 stocks A, B, C, D during the month of May.
The following data is the 5 day moving average (5 DMA) of these 5 stocks. 5 day moving average is the average prices of stocks on the last 5 days (including that day) when the stocks were traded in the stock exchange.
The following table gives the 5 DMA of stocks (in Rs.) at 1200 hours on 8 consecutive days when the markets were open.

Stock	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8
A	540	543	543	542	541	540	542	543
B	1020	1025	1024	1025	1024	1026	1028	1029
C	141	143	144	143	142	143	142	141
D	232	233	235	236	237	233	237	236
E	800	804	806	812	815	813	816	810

Assume all transactions happen at 1200 hours, and ignore all other costs involved in the transactions.

Q.40 [11594329]

If the price of C was Rs. 150 at 1200 hours on Day 3, what was the price (in Rs.) of C on Day 8?

1 ○ Rs. 147

2 ○ Rs. 145

3 ○ Rs. 141

4 ○ Rs.155

Solution:

Correct Answer : 2

Answer key/Solution

C3 + C4 + C5 + C6 + C7 = Rs. 142 × 5
C4 + C5 + C6 + C7 + C8 = Rs. 141 × 5
C3 – C8 = Rs. 5
So, C8 = Rs.145.

Bookmark FeedBack

Directions for questions 39 to 42: Answer the questions on the basis of the information given below.

Rakesh was a keen stock investor. During the Covid lockdown, since there was more time at hand he invested in 4 stocks A, B, C, D during the month of May.
The following data is the 5 day moving average (5 DMA) of these 5 stocks. 5 day moving average is the average prices of stocks on the last 5 days (including that day) when the stocks were traded in the stock exchange.
The following table gives the 5 DMA of stocks (in Rs.) at 1200 hours on 8 consecutive days when the markets were open.

Stock	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8
A	540	543	543	542	541	540	542	543
B	1020	1025	1024	1025	1024	1026	1028	1029
C	141	143	144	143	142	143	142	141
D	232	233	235	236	237	233	237	236
E	800	804	806	812	815	813	816	810

Assume all transactions happen at 1200 hours, and ignore all other costs involved in the transactions.

Q.41 [11594329]

If the 6 day moving average on Day 8 for B is Rs. 1,028, what is the price (in Rs.) of the stock on Day 8?

Solution:

Correct Answer : 1028

Answer key/Solution

Sum from B4 to B8 = 1029 × 5 = Rs. 5,145.
B3 to B8 = 1028 × 6 = Rs. 6,168
So, B3 = Rs. 1,023.
Also, sum from B4 to B8 = 1029 × 5
Sum from B3 to B7 = 1028 × 5
Hence, B8 – B3 = Rs. 5
So, B8 = 1023 + 5 = Rs. 1,028.

Bookmark FeedBack

Directions for questions 39 to 42: Answer the questions on the basis of the information given below.

Rakesh was a keen stock investor. During the Covid lockdown, since there was more time at hand he invested in 4 stocks A, B, C, D during the month of May.
The following data is the 5 day moving average (5 DMA) of these 5 stocks. 5 day moving average is the average prices of stocks on the last 5 days (including that day) when the stocks were traded in the stock exchange.
The following table gives the 5 DMA of stocks (in Rs.) at 1200 hours on 8 consecutive days when the markets were open.

Stock	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8
A	540	543	543	542	541	540	542	543
B	1020	1025	1024	1025	1024	1026	1028	1029
C	141	143	144	143	142	143	142	141
D	232	233	235	236	237	233	237	236
E	800	804	806	812	815	813	816	810

Assume all transactions happen at 1200 hours, and ignore all other costs involved in the transactions.

Q.42 [11594329]
Rakesh bought 100 stocks of E on each of the days from Day 3 to Day 7. He sold all of them at Rs. 820.10. The return on investment is closest to

1 ○ 1 %

2 ○ 0.5 %

3 ○ 0.9 %

4 ○ 1.25 %

Solution:
Correct Answer : 2

Answer key/Solution

Since he bought the same number of stocks on all the days from Day 3 to Day 7, the average cost price = 5 DMA on day 7.
The 5 DMA on day 7 for E is Rs. 816. Hence, profit = Rs. 4.10 which is approximately 0.5% of Rs. 816.

Bookmark FeedBack

Directions for questions 43 to 46: Answer the questions on the basis of the information given below.

Ravi, the coach of Indian Cricket Team, needs to select the eleven players for the match against Australia. Five players – Virat, Rohit, Hardik, Bumrah, and Rahul are certain to feature in the playing eleven. Out of the remaining 5 batsmen – Pant, Shikhar, Iyer, Krunal, and Kartik; and out of the remaining 5 bowlers – Shami, Bhuvi, Jadeja, Kuldeep, and Chahal; he needs to select 6 more players, subject to the following conditions:

(i) Out of Iyer, Pant, and Kartik, exactly one player must be selected.
(ii) Krunal and Jadeja cannot be in the team together, while Chahal and Shikhar cannot be in the team together.
(iii) If Shami is selected, then Shikhar must be selected.
(iv) If Kartik is selected, then Shami must be selected.
(v) If Chahal is selected, then at least 3 more of the remaining 4 bowlers must be selected.
(vi) Pant will be selected if and only if both Shikhar and Jadeja are selected.

Q.43 [11594329]
In how many ways can Ravi select the playing eleven?

Solution:
Correct Answer : 3

Answer key/Solution

From statement (v), if Chahal is selected, then at least 3 more of the 4 remaining bowlers – Shami, Bhuvi, Jadeja, and Kuldeep must be selected.
But from statement (iii), if Shami is selected, then Shikhar must be selected and then it will violate the condition given in statement (ii) i.e., Chahal and Shikhar cannot be in the team together. Hence, the 3 more bowlers that can be selected along with Chahal are Bhuvi, Jadeja, and Kuleep.
So from statement (ii), if Jadeja is selected, then Krunal cannot be selected. And, out of Iyer, Pant, and Kartik, exactly 1 is selected.
From statement (iv), if Shami is not selected, then Kartik cannot be selected. So along with Chahal, Bhuvi, Jadeja, and Kuleep, two more players need to be selected. But both Krunal and Shikhar are ruled out and out of Iyer, Pant, and Kartik, only one and that two other than Kartik must be selected.
Hence, if Chahal is selected, then it is not possible to select the total of 6 players. Therefore, in any possible selection of the six players, Chahal cannot be selected.

Case 1	Case 2	Case 3	Case 4	Case 5
Chahal	Chahal	Chahal	Chahal	Chahal
Shami	Shami	Shami	Bhuvi	Shami
Bhuvi	Bhuvi	Jadeja	Jadeja	Bhuvi
Jadeja	Kuldeep	Kuldeep	Kuldeep	Jadeja
× Shikhar	× Shikhar	× Shikhar	× Shikhar	Kuldeep
× Krunal	Krunal	× Krunal	× Krunal	× Shikhar
				× Krunal
Iyer/× Pant/Kartik	Iyer/× Pant/Kartik	Iyer/× Pant/Kartik	Iyer/× Pant/×Kartik	Iyer/× Pant/Kartik
Not Possible	Not Possible	Not Possible	Not Possible	Not Possible

Now, since out of Iyer, Pant and Kartik, exactly one player must be selected, that means that out of the remaining six players – Shikhar, Krunal, Shami, Buvi, Jadeja, and Kuldeep, five players must be selected. Further, it is given that Krunal and Jadeja cannot be in the team together therefore, the remaining 4 of the above players i.e., Shikhar, Shami, Bhuvi, and Kuldeep must be selected along with one out of Krunal and Jadeja.
So, the following combinations are possible:

(a)	(b)	(c)
Shikhar	Shikhar	Shikhar
Shami	Shami	Shami
Bhuvi	Bhuvi	Bhuvi
Kuldeep	Kuldeep	Kuldeep
Iyer	Kartik	Pant
Krunal	Krunal	Jadeja

Note: From statement (vi), if Pant is selected, then Shikhar and Jadeja both must be selected and vice versa. Therefore, either with Iyer or with Kartik, we are selecting Krunal but not Jadeja.

In 3 ways Ravi can select the playing eleven.

Bookmark FeedBack

Directions for questions 43 to 46: Answer the questions on the basis of the information given below.

Ravi, the coach of Indian Cricket Team, needs to select the eleven players for the match against Australia. Five players – Virat, Rohit, Hardik, Bumrah, and Rahul are certain to feature in the playing eleven. Out of the remaining 5 batsmen – Pant, Shikhar, Iyer, Krunal, and Kartik; and out of the remaining 5 bowlers – Shami, Bhuvi, Jadeja, Kuldeep, and Chahal; he needs to select 6 more players, subject to the following conditions:

(i) Out of Iyer, Pant, and Kartik, exactly one player must be selected.
(ii) Krunal and Jadeja cannot be in the team together, while Chahal and Shikhar cannot be in the team together.
(iii) If Shami is selected, then Shikhar must be selected.
(iv) If Kartik is selected, then Shami must be selected.
(v) If Chahal is selected, then at least 3 more of the remaining 4 bowlers must be selected.
(vi) Pant will be selected if and only if both Shikhar and Jadeja are selected.

Q.44 [11594329]
Who among the following players can never be selected for the playing eleven?

1 ○ Chahal

2 ○ Pant

3 ○ Krunal

4 ○ Jadeja

Solution:
Correct Answer : 1

Answer key/Solution

From statement (v), if Chahal is selected, then at least 3 more of the 4 remaining bowlers – Shami, Bhuvi, Jadeja, and Kuldeep must be selected.
But from statement (iii), if Shami is selected, then Shikhar must be selected and then it will violate the condition given in statement (ii) i.e., Chahal and Shikhar cannot be in the team together. Hence, the 3 more bowlers that can be selected along with Chahal are Bhuvi, Jadeja, and Kuleep.
So from statement (ii), if Jadeja is selected, then Krunal cannot be selected. And, out of Iyer, Pant, and Kartik, exactly 1 is selected.
From statement (iv), if Shami is not selected, then Kartik cannot be selected. So along with Chahal, Bhuvi, Jadeja, and Kuleep, two more players need to be selected. But both Krunal and Shikhar are ruled out and out of Iyer, Pant, and Kartik, only one and that two other than Kartik must be selected.
Hence, if Chahal is selected, then it is not possible to select the total of 6 players. Therefore, in any possible selection of the six players, Chahal cannot be selected.

Case 1	Case 2	Case 3	Case 4	Case 5
Chahal	Chahal	Chahal	Chahal	Chahal
Shami	Shami	Shami	Bhuvi	Shami
Bhuvi	Bhuvi	Jadeja	Jadeja	Bhuvi
Jadeja	Kuldeep	Kuldeep	Kuldeep	Jadeja
× Shikhar	× Shikhar	× Shikhar	× Shikhar	Kuldeep
× Krunal	Krunal	× Krunal	× Krunal	× Shikhar
				× Krunal
Iyer/× Pant/Kartik	Iyer/× Pant/Kartik	Iyer/× Pant/Kartik	Iyer/× Pant/×Kartik	Iyer/× Pant/Kartik
Not Possible	Not Possible	Not Possible	Not Possible	Not Possible

Now, since out of Iyer, Pant and Kartik, exactly one player must be selected, that means that out of the remaining six players – Shikhar, Krunal, Shami, Buvi, Jadeja, and Kuldeep, five players must be selected. Further, it is given that Krunal and Jadeja cannot be in the team together therefore, the remaining 4 of the above players i.e., Shikhar, Shami, Bhuvi, and Kuldeep must be selected along with one out of Krunal and Jadeja.
So, the following combinations are possible:

(a)	(b)	(c)
Shikhar	Shikhar	Shikhar
Shami	Shami	Shami
Bhuvi	Bhuvi	Bhuvi
Kuldeep	Kuldeep	Kuldeep
Iyer	Kartik	Pant
Krunal	Krunal	Jadeja

Note: From statement (vi), if Pant is selected, then Shikhar and Jadeja both must be selected and vice versa. Therefore, either with Iyer or with Kartik, we are selecting Krunal but not Jadeja.

Chahal can never be selected for the playing eleven.

Bookmark FeedBack

Directions for questions 43 to 46: Answer the questions on the basis of the information given below.

Ravi, the coach of Indian Cricket Team, needs to select the eleven players for the match against Australia. Five players – Virat, Rohit, Hardik, Bumrah, and Rahul are certain to feature in the playing eleven. Out of the remaining 5 batsmen – Pant, Shikhar, Iyer, Krunal, and Kartik; and out of the remaining 5 bowlers – Shami, Bhuvi, Jadeja, Kuldeep, and Chahal; he needs to select 6 more players, subject to the following conditions:

(i) Out of Iyer, Pant, and Kartik, exactly one player must be selected.
(ii) Krunal and Jadeja cannot be in the team together, while Chahal and Shikhar cannot be in the team together.
(iii) If Shami is selected, then Shikhar must be selected.
(iv) If Kartik is selected, then Shami must be selected.
(v) If Chahal is selected, then at least 3 more of the remaining 4 bowlers must be selected.
(vi) Pant will be selected if and only if both Shikhar and Jadeja are selected.

Q.45 [11594329]
How many of the remaining 5 batsmen will always be selected?

1 ○ 1

2 ○ 2

3 ○ 3

4 ○ 4

Solution:
Correct Answer : 1

Answer key/Solution

From statement (v), if Chahal is selected, then at least 3 more of the 4 remaining bowlers – Shami, Bhuvi, Jadeja, and Kuldeep must be selected.
But from statement (iii), if Shami is selected, then Shikhar must be selected and then it will violate the condition given in statement (ii) i.e., Chahal and Shikhar cannot be in the team together. Hence, the 3 more bowlers that can be selected along with Chahal are Bhuvi, Jadeja, and Kuleep.
So from statement (ii), if Jadeja is selected, then Krunal cannot be selected. And, out of Iyer, Pant, and Kartik, exactly 1 is selected.
From statement (iv), if Shami is not selected, then Kartik cannot be selected. So along with Chahal, Bhuvi, Jadeja, and Kuleep, two more players need to be selected. But both Krunal and Shikhar are ruled out and out of Iyer, Pant, and Kartik, only one and that two other than Kartik must be selected.
Hence, if Chahal is selected, then it is not possible to select the total of 6 players. Therefore, in any possible selection of the six players, Chahal cannot be selected.

Case 1	Case 2	Case 3	Case 4	Case 5
Chahal	Chahal	Chahal	Chahal	Chahal
Shami	Shami	Shami	Bhuvi	Shami
Bhuvi	Bhuvi	Jadeja	Jadeja	Bhuvi
Jadeja	Kuldeep	Kuldeep	Kuldeep	Jadeja
× Shikhar	× Shikhar	× Shikhar	× Shikhar	Kuldeep
× Krunal	Krunal	× Krunal	× Krunal	× Shikhar
				× Krunal
Iyer/× Pant/Kartik	Iyer/× Pant/Kartik	Iyer/× Pant/Kartik	Iyer/× Pant/×Kartik	Iyer/× Pant/Kartik
Not Possible	Not Possible	Not Possible	Not Possible	Not Possible

Now, since out of Iyer, Pant and Kartik, exactly one player must be selected, that means that out of the remaining six players – Shikhar, Krunal, Shami, Buvi, Jadeja, and Kuldeep, five players must be selected. Further, it is given that Krunal and Jadeja cannot be in the team together therefore, the remaining 4 of the above players i.e., Shikhar, Shami, Bhuvi, and Kuldeep must be selected along with one out of Krunal and Jadeja.
So, the following combinations are possible:

(a)	(b)	(c)
Shikhar	Shikhar	Shikhar
Shami	Shami	Shami
Bhuvi	Bhuvi	Bhuvi
Kuldeep	Kuldeep	Kuldeep
Iyer	Kartik	Pant
Krunal	Krunal	Jadeja

Note: From statement (vi), if Pant is selected, then Shikhar and Jadeja both must be selected and vice versa. Therefore, either with Iyer or with Kartik, we are selecting Krunal but not Jadeja.

One of the remaining 5 batsmen, i.e., Shikhar will always be selected.

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Directions for questions 43 to 46: Answer the questions on the basis of the information given below.

Ravi, the coach of Indian Cricket Team, needs to select the eleven players for the match against Australia. Five players – Virat, Rohit, Hardik, Bumrah, and Rahul are certain to feature in the playing eleven. Out of the remaining 5 batsmen – Pant, Shikhar, Iyer, Krunal, and Kartik; and out of the remaining 5 bowlers – Shami, Bhuvi, Jadeja, Kuldeep, and Chahal; he needs to select 6 more players, subject to the following conditions:

(i) Out of Iyer, Pant, and Kartik, exactly one player must be selected.
(ii) Krunal and Jadeja cannot be in the team together, while Chahal and Shikhar cannot be in the team together.
(iii) If Shami is selected, then Shikhar must be selected.
(iv) If Kartik is selected, then Shami must be selected.
(v) If Chahal is selected, then at least 3 more of the remaining 4 bowlers must be selected.
(vi) Pant will be selected if and only if both Shikhar and Jadeja are selected.

Q.46 [11594329]

In how many ways can Ravi select the playing eleven if Krunal is selected?

Solution:
Correct Answer : 2

Answer key/Solution

From statement (v), if Chahal is selected, then at least 3 more of the 4 remaining bowlers – Shami, Bhuvi, Jadeja, and Kuldeep must be selected.
But from statement (iii), if Shami is selected, then Shikhar must be selected and then it will violate the condition given in statement (ii) i.e., Chahal and Shikhar cannot be in the team together. Hence, the 3 more bowlers that can be selected along with Chahal are Bhuvi, Jadeja, and Kuleep.
So from statement (ii), if Jadeja is selected, then Krunal cannot be selected. And, out of Iyer, Pant, and Kartik, exactly 1 is selected.
From statement (iv), if Shami is not selected, then Kartik cannot be selected. So along with Chahal, Bhuvi, Jadeja, and Kuleep, two more players need to be selected. But both Krunal and Shikhar are ruled out and out of Iyer, Pant, and Kartik, only one and that two other than Kartik must be selected.
Hence, if Chahal is selected, then it is not possible to select the total of 6 players. Therefore, in any possible selection of the six players, Chahal cannot be selected.

Case 1	Case 2	Case 3	Case 4	Case 5
Chahal	Chahal	Chahal	Chahal	Chahal
Shami	Shami	Shami	Bhuvi	Shami
Bhuvi	Bhuvi	Jadeja	Jadeja	Bhuvi
Jadeja	Kuldeep	Kuldeep	Kuldeep	Jadeja
× Shikhar	× Shikhar	× Shikhar	× Shikhar	Kuldeep
× Krunal	Krunal	× Krunal	× Krunal	× Shikhar
				× Krunal
Iyer/× Pant/Kartik	Iyer/× Pant/Kartik	Iyer/× Pant/Kartik	Iyer/× Pant/×Kartik	Iyer/× Pant/Kartik
Not Possible	Not Possible	Not Possible	Not Possible	Not Possible

Now, since out of Iyer, Pant and Kartik, exactly one player must be selected, that means that out of the remaining six players – Shikhar, Krunal, Shami, Buvi, Jadeja, and Kuldeep, five players must be selected. Further, it is given that Krunal and Jadeja cannot be in the team together therefore, the remaining 4 of the above players i.e., Shikhar, Shami, Bhuvi, and Kuldeep must be selected along with one out of Krunal and Jadeja.
So, the following combinations are possible:

(a)	(b)	(c)
Shikhar	Shikhar	Shikhar
Shami	Shami	Shami
Bhuvi	Bhuvi	Bhuvi
Kuldeep	Kuldeep	Kuldeep
Iyer	Kartik	Pant
Krunal	Krunal	Jadeja

Note: From statement (vi), if Pant is selected, then Shikhar and Jadeja both must be selected and vice versa. Therefore, either with Iyer or with Kartik, we are selecting Krunal but not Jadeja.

Ravi can select the playing eleven if Krunal is selected in two ways.

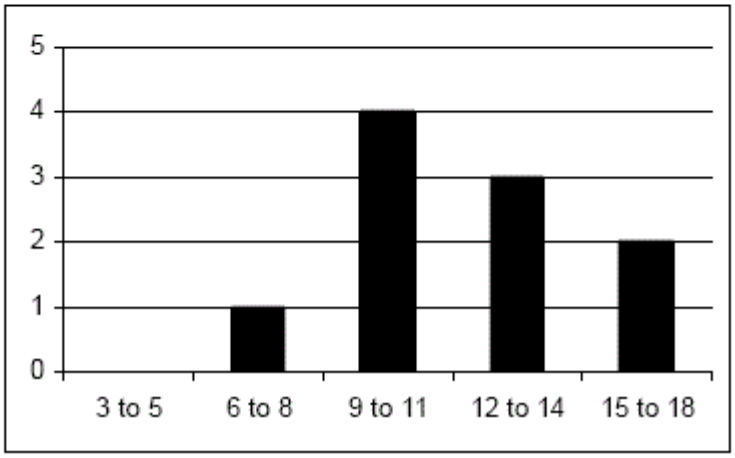
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Directions for questions 47 to 50: Answer the questions on the basis of the information given below.

Ten players P – 1 through P – 10 play a game of rolling a dice. The game has three rounds. In the first round, each player rolls the dice once. Similarly, all the ten players roll the dice once each in the second and the third round. The dice is a standard dice having six faces numbered 1 through 6.
The score of a player in a round is the number displayed on the dice, when he/she rolled the dice in that particular round. The total score of a player is the sum of scores in the three rounds. The total scores in Round – 1, Round – 2 and Round – 3 were 45, 37 and 35 respectively.
The table below gives the score of the ten players in the three rounds. Some of the values of scores are missing.

	Round 1	Round 2	Round 3
P-1	6		1
P-2		3	5
P-3	4		
P-4	5	4	1
P-5		2	
P-6	6		1
P-7	4	1	2
P-8			4
P-9	1	3	
P-10		4	4

The histogram below displays the number of players whose total score was in the given range.
For example, 3 players had a total score between 12 and 14, both inclusive.



Additionally,
(i) P – 4 and P – 5 had identical total scores.
(ii) Considering all the rounds of all the players, the frequency of the score '5' was an even number.
(iii) The frequencies of the score '6' in Round – 3, Round – 2 and Round – 1 form an increasing Arithmetic Progression.
(iv) Exactly five players scored distinct scores in all the three rounds they played.
(v) The total score of P – 1 was more than that of P – 6.

Q.47 [11594329]
For how many of the ten players can the scores in their three rounds be uniquely determined?

1 ○ 8

2 ○ 7

3 ○ 5

4 ○ 6

Solution:
Correct Answer : 4

Answer key/Solution

As per the histogram:

Total Score	No. of players
3 to 5	0
6 to 8	1
9 to 11	4
12 to 15	3
15 to 18	2

Rewriting the table with the total scores of each round:

w	Round 1	Round 2	Round 3
P1	6		1
P2		3	5
P3	4		
P4	5	4	1
P5		2	
P6	6		1
P7	4	1	2
P8			4
P9	1	3	
P10		4	4
Total	45	37	35

Using statement (i),

All distinct?		Round 1	Round 2	Round 3	Total	Category
	P1	6		1		
	P2		3	5		
	P3	4				
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
	P6	6		1		
Yes	P7	4	1	2	7	6 to 8

	P8			4		
	P9	1	3			9 to 11
	P10		4	4		
	Total	45	37	35		

Further, using that score in each round is between 1 and 6, we can conclude that only P – 3 and P – 8 can have a score in the range **'15 to 18'.**

All distinct?		Round 1	Round 2	Round 3	Total	Category
	P1	6		1		
	P2		3	5		
	P3	4				15 to 18
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
	P6	6		1		
Yes	P7	4	1	2	7	6 to 8
	P8			4		15 to 18
	P9	1	3			9 to 11
	P10		4	4		
	Total	45	37	35		

Total score of Round 3 is 35.
Thus, the sum of scores of P – 3, P – 5 and P – 9 in Round 3 is 17.
The only way to get a total of 17 is by **adding 6, 6 and 5.**
This means Round – 3 has two scores of 6.
Using statement (iii), and knowing that round 2 cannot have four scores of 6:
we get that **Round 3, Round 2, and Round 1 have 2, 3, and 4 scores of 6 respectively.**
Thus, the unknown scores of **Round 2 are 6, 6, 6 and 2.**
For Round 1, the unknown scores are either 6, 6, 5, 2 or 6, 6, 4, 3.
But, using statement (ii), only possibility for the unknown scores of **Round 1 is 6, 6, 5, 2.**
Using statement (v), the table can be filled further:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
	P2		3	5		12 to 14
	P3	4	6			15 to 18
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
	P8		6	4		15 to 18
	P9	1	3			9 to 11
	P10		4	4		12 to 14
	Total	45	37	35		

Now considering Round 1's unknown score of 6, 6, 5, 2, P – 5 must have scored a 2 in Round – 1.

All distinct?		Round 1	Round 2	Round 3	Total	Category

No	P1	6	6	1	13	12 to 14
	P2		3	5		12 to 14
	P3	4	6			15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
	P8		6	4		15 to 18
No	P9	1	3			9 to 11
No	P10		4	4		12 to 14
	Total	45	37	35		

Finally, using statement (iv), we have **two possibilities**:
Possibility 1:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
Yes	P2	6	3	5	14	12 to 14
No	P3	4	6	6	16	15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
No	P8	6	6	4	16	15 to 18
Yes	P9	1	3	5	9	9 to 11
No	P10	5	4	4	13	12 to 14
	Total	45	37	35		

Possibility 2:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
No	P2	5	3	5	13	12 to 14
Yes	P3	4	6	5	15	15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
No	P8	6	6	4	16	15 to 18
Yes	P9	1	3	6	10	9 to 11
No	P10	6	4	4	14	12 to 14
	Total	45	37	35		

For only six players, we can determine uniquely their scores in all the three rounds i.e., P – 1, P – 4, P – 5, P – 6, P – 7, P – 8.

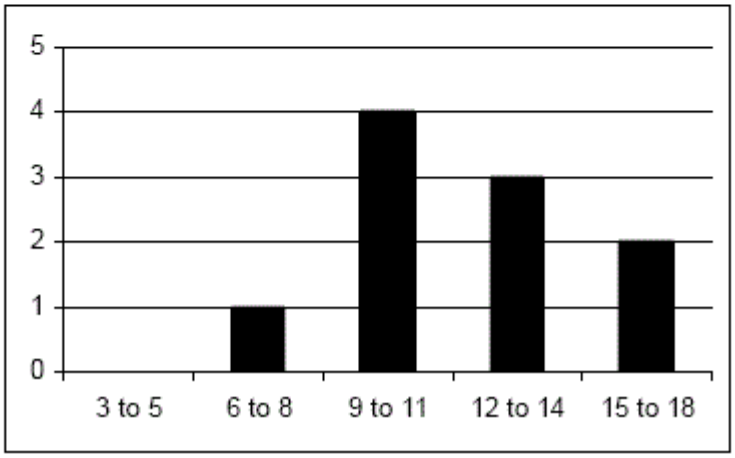
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Directions for questions 47 to 50: Answer the questions on the basis of the information given below.

Ten players P – 1 through P – 10 play a game of rolling a dice. The game has three rounds. In the first round, each player rolls the dice once. Similarly, all the ten players roll the dice once each in the second and the third round. The dice is a standard dice having six faces numbered 1 through 6.
The score of a player in a round is the number displayed on the dice, when he/she rolled the dice in that particular round. The total score of a player is the sum of scores in the three rounds. The total scores in Round – 1, Round – 2 and Round – 3 were 45, 37 and 35 respectively.
The table below gives the score of the ten players in the three rounds. Some of the values of scores are missing.

	Round 1	Round 2	Round 3
P-1	6		1
P-2		3	5
P-3	4		
P-4	5	4	1
P-5		2	
P-6	6		1
P-7	4	1	2
P-8			4
P-9	1	3	
P-10		4	4

The histogram below displays the number of players whose total score was in the given range.
For example, 3 players had a total score between 12 and 14, both inclusive.



Additionally,
(i) P – 4 and P – 5 had identical total scores.
(ii) Considering all the rounds of all the players, the frequency of the score '5' was an even number.
(iii) The frequencies of the score '6' in Round – 3, Round – 2 and Round – 1 form an increasing Arithmetic Progression.
(iv) Exactly five players scored distinct scores in all the three rounds they played.
(v) The total score of P – 1 was more than that of P – 6.

Q.48 [11594329]
Which of the following statements MUST be FALSE?

1 ○ The total score of P – 3 was more than that of P – 8

2 ○ The total score of P – 6 was less than that of P – 9

3 ○ The total score of P – 10 was less than that of P – 8

4 ○ The total scores of P – 1 and P – 2 were equal

Solution:
Correct Answer : 1

Answer key/Solution

As per the histogram:

Total Score	No. of players
3 to 5	0
6 to 8	1
9 to 11	4
12 to 15	3
15 to 18	2

Rewriting the table with the total scores of each round:

w	Round 1	Round 2	Round 3
P1	6		1
P2		3	5
P3	4		
P4	5	4	1
P5		2	
P6	6		1
P7	4	1	2
P8			4
P9	1	3	
P10		4	4
Total	45	37	35

Using statement (i),

All distinct?		Round 1	Round 2	Round 3	Total	Category
	P1	6		1		
	P2		3	5		
	P3	4				
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
	P6	6		1		
Yes	P7	4	1	2	7	6 to 8

	P8			4		
	P9	1	3			9 to 11
	P10		4	4		
	Total	45	37	35		

Further, using that score in each round is between 1 and 6, we can conclude that only P – 3 and P – 8 can have a score in the range **'15 to 18'.**

All distinct?		Round 1	Round 2	Round 3	Total	Category
	P1	6		1		
	P2		3	5		
	P3	4				15 to 18
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
	P6	6		1		
Yes	P7	4	1	2	7	6 to 8
	P8			4		15 to 18
	P9	1	3			9 to 11
	P10		4	4		
	Total	45	37	35		

Total score of Round 3 is 35.
Thus, the sum of scores of P – 3, P – 5 and P – 9 in Round 3 is 17.
The only way to get a total of 17 is by **adding 6, 6 and 5.**
This means Round – 3 has two scores of 6.
Using statement (iii), and knowing that round 2 cannot have four scores of 6:
we get that **Round 3, Round 2, and Round 1 have 2, 3, and 4 scores of 6 respectively.**
Thus, the unknown scores of **Round 2 are 6, 6, 6 and 2.**
For Round 1, the unknown scores are either 6, 6, 5, 2 or 6, 6, 4, 3.
But, using statement (ii), only possibility for the unknown scores of **Round 1 is 6, 6, 5, 2.**
Using statement (v), the table can be filled further:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
	P2		3	5		12 to 14
	P3	4	6			15 to 18
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
	P8		6	4		15 to 18
	P9	1	3			9 to 11
	P10		4	4		12 to 14
	Total	45	37	35		

Now considering Round 1's unknown score of 6, 6, 5, 2, P – 5 must have scored a 2 in Round – 1.

All distinct?		Round 1	Round 2	Round 3	Total	Category

No	P1	6	6	1	13	12 to 14
	P2		3	5		12 to 14
	P3	4	6			15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
	P8		6	4		15 to 18
No	P9	1	3			9 to 11
No	P10		4	4		12 to 14
	Total	45	37	35		

Finally, using statement (iv), we have **two possibilities**:
Possibility 1:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
Yes	P2	6	3	5	14	12 to 14
No	P3	4	6	6	16	15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
No	P8	6	6	4	16	15 to 18
Yes	P9	1	3	5	9	9 to 11
No	P10	5	4	4	13	12 to 14
	Total	45	37	35		

Possibility 2:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
No	P2	5	3	5	13	12 to 14
Yes	P3	4	6	5	15	15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
No	P8	6	6	4	16	15 to 18
Yes	P9	1	3	6	10	9 to 11
No	P10	6	4	4	14	12 to 14
	Total	45	37	35		

Option 1 is definitely false.

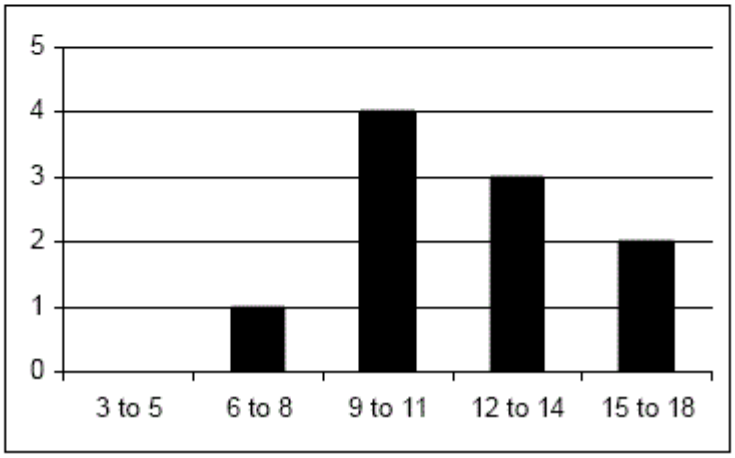
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The table below gives the score of the ten players in the three rounds. Some of the values of scores are missing.

	Round 1	Round 2	Round 3
P-1	6		1
P-2		3	5
P-3	4		
P-4	5	4	1
P-5		2	
P-6	6		1
P-7	4	1	2
P-8			4
P-9	1	3	
P-10		4	4

The histogram below displays the number of players whose total score was in the given range.
For example, 3 players had a total score between 12 and 14, both inclusive.



Additionally,
(i) P – 4 and P – 5 had identical total scores.
(ii) Considering all the rounds of all the players, the frequency of the score '5' was an even number.
(iii) The frequencies of the score '6' in Round – 3, Round – 2 and Round – 1 form an increasing Arithmetic Progression.
(iv) Exactly five players scored distinct scores in all the three rounds they played.
(v) The total score of P – 1 was more than that of P – 6.

Q.49 [11594329]
Considering all the rounds of all the players, what was the frequency of the score '5'?

Solution:

Answer key/Solution

Correct Answer : 4

As per the histogram:

Total Score	No. of players
3 to 5	0
6 to 8	1
9 to 11	4
12 to 15	3
15 to 18	2

Rewriting the table with the total scores of each round:

w	Round 1	Round 2	Round 3
P1	6		1
P2		3	5
P3	4		
P4	5	4	1
P5		2	
P6	6		1
P7	4	1	2
P8			4
P9	1	3	
P10		4	4
Total	45	37	35

Using statement (i),

All distinct?		Round 1	Round 2	Round 3	Total	Category
	P1	6		1		
	P2		3	5		
	P3	4				
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
	P6	6		1		
Yes	P7	4	1	2	7	6 to 8
	P8			4		
	P9	1	3			9 to 11
	P10		4	4		
	Total	45	37	35		

Further, using that score in each round is between 1 and 6, we can conclude that only P – 3 and P – 8 can have a score in the range **'15 to 18'.**

All distinct?		Round 1	Round 2	Round 3	Total	Category
	P1	6		1		

	P2		3	5		
	P3	4				15 to 18
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
	P6	6		1		
Yes	P7	4	1	2	7	6 to 8
	P8			4		15 to 18
	P9	1	3			9 to 11
	P10		4	4		
	Total	45	37	35		

Total score of Round 3 is 35.
Thus, the sum of scores of P – 3, P – 5 and P – 9 in Round 3 is 17.
The only way to get a total of 17 is by **adding 6, 6 and 5.**
This means Round – 3 has two scores of 6.
Using statement (iii), and knowing that round 2 cannot have four scores of 6:
we get that **Round 3, Round 2, and Round 1 have 2, 3, and 4 scores of 6 respectively.**
Thus, the unknown scores of **Round 2 are 6, 6, 6 and 2.**
For Round 1, the unknown scores are either 6, 6, 5, 2 or 6, 6, 4, 3.
But, using statement (ii), only possibility for the unknown scores of **Round 1 is 6, 6, 5, 2.**
Using statement (v), the table can be filled further:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
	P2		3	5		12 to 14
	P3	4	6			15 to 18
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
	P8		6	4		15 to 18
	P9	1	3			9 to 11
	P10		4	4		12 to 14
	Total	45	37	35		

Now considering Round 1's unknown score of 6, 6, 5, 2, P – 5 must have scored a 2 in Round – 1.

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
	P2		3	5		12 to 14
	P3	4	6			15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8

	P8		6	4		15 to 18
No	P9	1	3			9 to 11
No	P10		4	4		12 to 14
	Total	45	37	35		

Finally, using statement (iv), we have **two possibilities:**
Possibility 1:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
Yes	P2	6	3	5	14	12 to 14
No	P3	4	6	6	16	15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
No	P8	6	6	4	16	15 to 18
Yes	P9	1	3	5	9	9 to 11
No	P10	5	4	4	13	12 to 14
	Total	45	37	35		

Possibility 2:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
No	P2	5	3	5	13	12 to 14
Yes	P3	4	6	5	15	15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
No	P8	6	6	4	16	15 to 18
Yes	P9	1	3	6	10	9 to 11
No	P10	6	4	4	14	12 to 14
	Total	45	37	35		

Considering all the rounds of all the players, the frequency of the score '5' is 4.

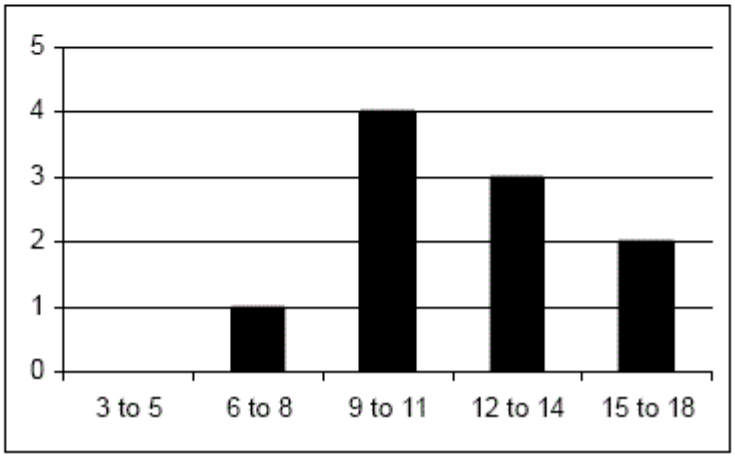
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Directions for questions 47 to 50: Answer the questions on the basis of the information given below.

Ten players P – 1 through P – 10 play a game of rolling a dice. The game has three rounds. In the first round, each player rolls the dice once. Similarly, all the ten players roll the dice once each in the second and the third round. The dice is a standard dice having six faces numbered 1 through 6.
The score of a player in a round is the number displayed on the dice, when he/she rolled the dice in that particular round. The total score of a player is the sum of scores in the three rounds. The total scores in Round – 1, Round – 2 and Round – 3 were 45, 37 and 35 respectively.
The table below gives the score of the ten players in the three rounds. Some of the values of scores are missing.

	Round 1	Round 2	Round 3
P-1	6		1
P-2		3	5
P-3	4		
P-4	5	4	1
P-5		2	
P-6	6		1
P-7	4	1	2
P-8			4
P-9	1	3	
P-10		4	4

The histogram below displays the number of players whose total score was in the given range.
For example, 3 players had a total score between 12 and 14, both inclusive.



Additionally,
(i) P – 4 and P – 5 had identical total scores.
(ii) Considering all the rounds of all the players, the frequency of the score '5' was an even number.
(iii) The frequencies of the score '6' in Round – 3, Round – 2 and Round – 1 form an increasing Arithmetic Progression.
(iv) Exactly five players scored distinct scores in all the three rounds they played.
(v) The total score of P – 1 was more than that of P – 6.

Q.50 [11594329]
If P – 10 had a score of 5 in Round – 1, then what was the total score of P – 3?

1 ○ 14

2 ○ 15

3 ○ 16

4 ○ 15 or 16

Solution:
Correct Answer : 3

Answer key/Solution

As per the histogram:

Total Score	No. of players
3 to 5	0
6 to 8	1
9 to 11	4
12 to 15	3
15 to 18	2

Rewriting the table with the total scores of each round:

w	Round 1	Round 2	Round 3
P1	6		1
P2		3	5
P3	4		
P4	5	4	1
P5		2	
P6	6		1
P7	4	1	2
P8			4
P9	1	3	
P10		4	4
Total	45	37	35

Using statement (i),

All distinct?		Round 1	Round 2	Round 3	Total	Category
	P1	6		1		
	P2		3	5		
	P3	4				
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
	P6	6		1		
Yes	P7	4	1	2	7	6 to 8

	P8			4		
	P9	1	3			9 to 11
	P10		4	4		
	Total	45	37	35		

Further, using that score in each round is between 1 and 6, we can conclude that only P – 3 and P – 8 can have a score in the range **'15 to 18'.**

All distinct?		Round 1	Round 2	Round 3	Total	Category
	P1	6		1		
	P2		3	5		
	P3	4				15 to 18
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
	P6	6		1		
Yes	P7	4	1	2	7	6 to 8
	P8			4		15 to 18
	P9	1	3			9 to 11
	P10		4	4		
	Total	45	37	35		

Total score of Round 3 is 35.
Thus, the sum of scores of P – 3, P – 5 and P – 9 in Round 3 is 17.
The only way to get a total of 17 is by **adding 6, 6 and 5.**
This means Round – 3 has two scores of 6.
Using statement (iii), and knowing that round 2 cannot have four scores of 6:
we get that **Round 3, Round 2, and Round 1 have 2, 3, and 4 scores of 6 respectively.**
Thus, the unknown scores of **Round 2 are 6, 6, 6 and 2.**
For Round 1, the unknown scores are either 6, 6, 5, 2 or 6, 6, 4, 3.
But, using statement (ii), only possibility for the unknown scores of **Round 1 is 6, 6, 5, 2.**
Using statement (v), the table can be filled further:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
	P2		3	5		12 to 14
	P3	4	6			15 to 18
Yes	P4	5	4	1	10	9 to 11
	P5		2		10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
	P8		6	4		15 to 18
	P9	1	3			9 to 11
	P10		4	4		12 to 14
	Total	45	37	35		

Now considering Round 1's unknown score of 6, 6, 5, 2, P – 5 must have scored a 2 in Round – 1.

All distinct?		Round 1	Round 2	Round 3	Total	Category

No	P1	6	6	1	13	12 to 14
	P2		3	5		12 to 14
	P3	4	6			15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
	P8		6	4		15 to 18
No	P9	1	3			9 to 11
No	P10		4	4		12 to 14
	Total	45	37	35		

Finally, using statement (iv), we have **two possibilities**:
Possibility 1:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
Yes	P2	6	3	5	14	12 to 14
No	P3	4	6	6	16	15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
No	P8	6	6	4	16	15 to 18
Yes	P9	1	3	5	9	9 to 11
No	P10	5	4	4	13	12 to 14
	Total	45	37	35		

Possibility 2:

All distinct?		Round 1	Round 2	Round 3	Total	Category
No	P1	6	6	1	13	12 to 14
No	P2	5	3	5	13	12 to 14
Yes	P3	4	6	5	15	15 to 18
Yes	P4	5	4	1	10	9 to 11
No	P5	2	2	6	10	9 to 11
Yes	P6	6	2	1	9	9 to 11
Yes	P7	4	1	2	7	6 to 8
No	P8	6	6	4	16	15 to 18
Yes	P9	1	3	6	10	9 to 11
No	P10	6	4	4	14	12 to 14
	Total	45	37	35		

If P – 10 had a score of 5 in Round – 1, then the total score of P – 3 was 16.

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Sec 3

Q.51 [11594329]

(a, b, c, d) is an ordered quadruple whose elements are chosen from the numbers {0, 1, 2, 3}. If a, b, c, d are not necessarily distinct, find the number of ordered quadruples that can be chosen such that ab – cd is odd.

1 ○ 96

2 ○ 56

3 ○ 48

4 ○ 112

Solution:

Correct Answer : 1

Answer key/Solution

Given: ab – cd = odd number
So it is either "odd – even" or "even – odd".
ab = odd if both a and b are odd.
So, there are 4 ways i.e., (1, 1), (1, 3), (3, 1), (3, 3).
cd = even if atleast one of c and d is even.
(c, d) can have total of 4 × 4 – 4 = 12 combinations.
Hence, odd – even = 4 × 12 = 48 and even – odd = 48.
Hence, total = 96 cases.

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Q.52 [11594329]

How many integer points are there for which the polynomial $(x - 2^2)(x - 3^2)(x - 4^2)...(x - 11^2) \leq 0$?

1 ○ 117

2 ○ 118

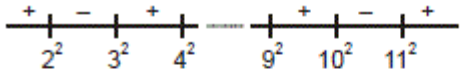
3 ○ 70

4 ○ 65

Solution:

Correct Answer : 3

Answer key/Solution



For x = 0, 10 solutions are possible, i.e., 2^2, 3^2, 4^2, ..., 11^2
For x < 0, $(11^2 - 10^2) - 1 = 20$,
$(5^2 - 4^2) - 1 = 8$,
$(9^2 - 8^2) - 1 = 16$,
$(3^2 - 2^2) - 1 = 4$
$(7^2 - 6^2) - 1 = 12$
Hence, total solutions = 10 + 20 + 8 + 16 + 4 + 12 = 70.

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Q.53 [11594329]

For two non-zero real numbers x and y, it is known that (x + y) = 4 and xy = −3. Find the value of the expression $x + y + x^5 + y^5$.

✖

Solution:

Correct Answer : 2168

Your Answer : 248

Answer key/Solution

$(x + y) + x^5 + y^5 = 4 + (x^3 + y^3)(x^2 + y^2) - (x^2y^2)(x + y)$
$= 4 + (x^3 + y^3)(x^2 + y^2) - 36$
$(x^3 + y^3) = (x + y)^3 - 3xy\,(x + y) = 64 + 9\,(4) = 100$
$(x^2 + y^2) = (x + y)^2 - 2xy = 16 + 6 = 22$
Hence, $4 + (x^3 + y^3)(x^2 + y^2) - 36 = 2200 - 32 = 2168$.

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Q.54 [11594329]

What is the smallest value of n for which $\frac{n!}{(n-4)!}$ has 3 trailing zeroes?

✖

Solution:
Correct Answer : 125
Your Answer : 10

Answer key/Solution

$$\frac{n!}{(n-4)!} = n \times (n-1) \times (n-2) \times (n-3)$$

The smallest value of n that has 5^3 in it is 125. 124 and 122 together contribute three 2's.
So, this expression has 3 trailing zeroes.

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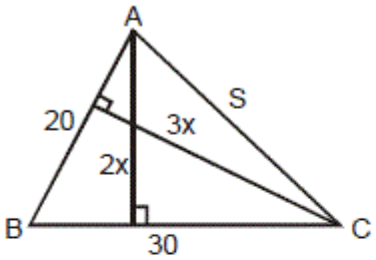
Q.55 [11594329]

The two sides of a triangle are 20 and 30 units. The length of altitude to the third side is the average of the lengths of the other two altitudes. The length of the third side is

✖

Solution:
Correct Answer : 24
Your Answer : 25

Answer key/Solution



Let the lengths of altitudes to sides 20 and 30 be h1 and h2 respectively.

So, area of triangle ABC = $\frac{1}{2} \times 20 \times h1 = \frac{1}{2} \times 30 \times h2$

$\Rightarrow h1 : h2 = 3 : 2$
Let h1 and h2 be 3x and 2x respectively.
So, the length of the third altitude will be 2.5x.

Therefore, area of triangle ABC = $\frac{1}{2} \times 20 \times 3x = \frac{1}{2} \times S \times 2.5x$ (Let S be the third side)

$\Rightarrow S = 24$ units.

Bookmark FeedBack

Q.56 [11594329]

6 people $P_1, P_2, P_3, \ldots, P_6$ have to be arranged in a straight line. P_k has to be next to one of or both of P_{k+1} and P_{k-1}. Find the number of ways they can be arranged.

1 ○ 12

2 ○ 28

3 ○ 30

4 ○ 54

Solution:
Correct Answer : 4

Answer key/Solution

P_1 will be next to P_2 and P_6 will be next to P_5 always.
Case 1: P_3 and P_4 are together.

$\boxed{P_1P_2}$ $\boxed{P_3P_4}$ $\boxed{P_5P_6}$ $= 3! \times (2!)^3 = 48$

Case 2: P_3 and P_4 are not together.

P_3 $\boxed{P_2P_1}$ P_4 $\boxed{P_5P_6}$ = 2 ways

$\boxed{P_1P_2}$ P_3 $\boxed{P_6P_5}$ P_4 = 2 ways

P_3 $\boxed{P_2P_1}$ $\boxed{P_6P_5}$ P_4 = 2 ways

Total ways = 48 + 2 × 3 = 54 ways.

Bookmark FeedBack

Q.57 [11594329]

A father wants to distribute a certain sum of money between his daughter and son in such a way that if both of them invest their shares in the scheme that offers compound interest at 20% per annum, for t and t + 2 years respectively, then the two shares grow to become equal. If the son's share was Rs. 4,000, then the total money distributed by the father was

1 ○ Rs. 5,760

2 ○ Rs. 2,778

3 ○ Rs. 6,778

4 ○ Rs. 9,760

Solution:
Correct Answer : 4

Answer key/Solution

Let the daughter get D amount of share.
Then, $D(1.2)^t = 4000\ (1.2)^{t+2}$;
$\Rightarrow D = 4000\ (1.2)^2$ = Rs. 5,760
So, total money distributed = 4000 + 5760 = Rs. 9,760.

Bookmark FeedBack

Q.58 [11594329]

A can complete a job working alone in 40 days. A works with B on day 1, With C on day 2, with B again on day 3, with C on day 4 … and so on. At the end of the 20th day, the job is completed. B does 30% of the work, C does 20% of the work. If they work on the job simultaneously the job gets completed on the

1 ○ 10th day

2 ○ 13th day

3 ○ 14th day

4 ○ 16th day

✖

Solution:
Correct Answer : 3
Your Answer : 1

Answer key/Solution

The pattern in which A, B and C works for 20 days is:

$\frac{A,B}{1}\ \frac{A,C}{2}\ \frac{A,B}{3}\ \frac{A,C}{4}\cdots$ so on.

Clearly, A works for 20 days, B works for 10 days, and C works for 10 days.
Therefore, A does 50% in 20 days, i.e., 2.5% in 1 day.
Similarly, B does 30% in 10 days, i.e., 3% in 1 day.
And, C does 2% in 1 day.
Together in 1 day they did 7.5% of work.

So it takes $\frac{100}{7.5}=13\frac{1}{3}$ days. i.e., almost 14 days to complete the entire work if they all work together.

Bookmark FeedBack

Q.59 [11594329]

Ratna's family of five (Ratna, her Spouse and 3 children) have to cross a river. The boat at the river bank is very small. It can accommodate 1 adult or a maximum of 2 kids. It takes 15 minutes to row across the river to the opposite bank. What is the least amount of time needed by the entire family to cross the river to the other river bank (in minutes)?

✖

Solution:
Correct Answer : 165
Your Answer : 60

Answer key/Solution

1st Trip : 2 children cross over.
2nd Trip : 1 child comes back (1 is on the opposite bank).
3rd Trip : 1 adult cross over.
4th trip : The other child from the opposite banks comes back alone (1 adult is on the opposite bank).
5th Trip : 2 children cross over.
6th trip : 1 child comes back (1 adult, 1 child is on the opposite bank).
7th Trip : 1 adult goes to the other bank.
8th Trip : 1 child comes back (2 adults is on the opposite bank).
9th Trip: 2 children row to the opposite bank.
10th Trip : 1 child comes back (1 child, 2 adults is on the opposite bank).
11th Trip : Both children cross over to the other Bank.
So, we need atleast 11 trips. So, in all it takes a minimum of 165 minutes for the family to cross over to the other side.

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Q.60 [11594329]

If (0, 0), (6, 2), (-2, 6) and (a, b) are the coordinates of the vertices of a parallelogram, what is the least possible value of a + b?

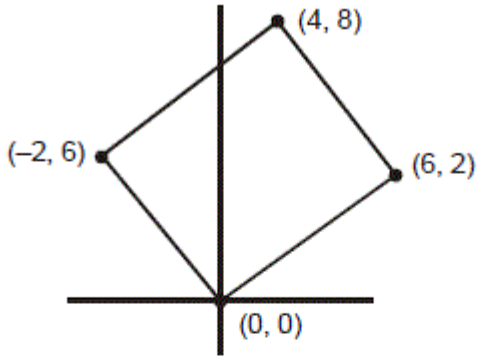
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Solution:
Correct Answer : -4
Your Answer : -2

Answer key/Solution

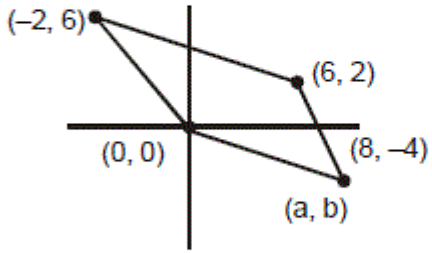
There are 3 possible solutions for the coordinates (a, b).

I.



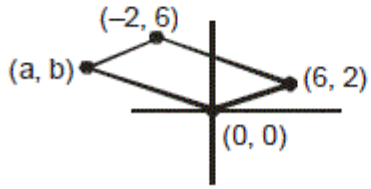
(a, b) = (4, 8)

II.



(a, b) = (8, -4)

III.



(a, b) = (–8, 4)

Hence, the least value of a + b = –8 + 4 = –4.

Bookmark FeedBack

Q.61 [11594329]

A fruit seller has three types of fruits in his shop - mango, orange, and guava. On a particular day, he sells 75 mangoes, 90 oranges and 120 guavas. If none of the customers bought more than three fruits of each type, then what is the minimum number of customers that must have visited the shop that day?

1 ○ 30

2 ○ 35

3 ○ 45

4 ○ 40

✔

Solution:
Correct Answer : 4
Your Answer : 4

Answer key/Solution

In order to minimise the number of customers, let us consider that initially 25 customers bought 3 fruits of each type.
So, now, all 75 mangoes are sold, and 90 – 75 = 15 oranges are remaining to be sold.
75 guavas are sold so remaining guavas = 120 – 75 = 45.
Now, another 5 customers bought 3 fruits of each type.
So, remaining all 15 oranges are sold and 15 guavas are sold
∴ Remaining guavas = 45 – 15 = 30
∴ In order to buy 30 guavas, 10 more customers must have visited the shop.
Hence, the minimum number of customers who visited the shop = 25 + 5 + 10 = 40.

Bookmark FeedBack

Q.62 [11594329]

Brinda and Anusha run in opposite directions on a circular track, starting at diametrically opposite points. Each girl runs at a constant speed. They first meet after Brinda has run 250 meters. They meet for the third time after Anusha has run 600 meters past their first meeting point. What is the length of the track (in meters)?

1 ○ 600

2 ○ 700

3 ○ 800

4 ○ 1000

Solution:
Correct Answer : 3

Answer key/Solution

When together they cover half a circle, they meet for the first time.
After this they have to cover 2 full circles to meet for the third time. This is 4 half circles.
So, Anusha must be covering 150 m when together they cover half a circle.
So, half a circle must be 250 + 150 = 400 m.
Hence, circumference = 800 m.

Bookmark FeedBack

Q.63 [11594329]

The sequence a_n is given by $a_1 = 1$, $a_2 = 3$ and $a_{n+2} = a_n + a_{n+1}$ for $n \geq 1$. How many of the first 2021 elements of the sequence are odd?

✖

Solution:
Correct Answer : 1348
Your Answer : 2020

Answer key/Solution

Given: $a_1 = 1$, $a_2 = 3$ and $a_{n+2} = a_n + a_{n+1}$.
$a_3 = 1 + 3 = 4$,
$a_4 = 3 + 4 = 7$,
$a_5 = 4 + 7 = 11$,
$a_6 = 7 + 11 = 18$ ... and so on.
The sequence would be (odd, odd, even, odd, odd, even,....).
So, in any group of 3 terms, 2 terms are odd and the third term is even.
So, 2021 when divided by 3 leaves a remainder of 2.

So, $2019\frac{2}{3} + 2 = 1348$ would be the number of odd terms.

Bookmark FeedBack

Q.64 [11594329]

Raghu and Ravi run 10 kilometers. They start at the same place, run 5 kilometers up a hill, and return to the starting point by the same route. Ravi has a 10-minute head start and runs at the rate of 15 km/hr uphill and 18 km/hr downhill. Raghu runs 12 km/hr uphill and 16 km/hr downhill. For how much time has Raghu travelled, when they pass in the opposite directions?

1 ○ 6 min

2 ○ 16 min

3 ○ 26 min

4 ○ 36 min

Solution:
Correct Answer : 2

Answer key/Solution

Let us assume that Ravi Starts at 9 : 00 AM.
He reaches the top of the hill at 9 : 20 AM.
Raghu starts at 9 : 10 AM. Raghu covers 2 km in 10 minutes.
Distance that separates them is 3 km.
Ravi is coming downhill at 18 km/hr. Ragu is going up at 12 km/hr.
Relative speed = 30 km/hr.
Therefore, it takes 6 minutes for them to meet.
So, Raghu has travelled for 16 minutes.

Bookmark FeedBack

Q.65 [11594329]

Richa's chemistry teacher asked the class to make a saltwater solution with a 5 : 7 ratio of salt to water. Richa misheard her teacher and made a 100 mL solution with a 1 : 3 ratio of water to salt instead. Which of the following will give the correct solution?

1 ○ Addition of 180 ml of water

2 ○ Addition of 80 ml of water

3 ○ Addition of about 27 gms of salt

4 ○ Addition of 28.5 gms of salt

✔

Solution:
Correct Answer : 2
Your Answer : 2

Answer key/Solution

100 ml of 1 : 3 water to salt solution consists of 75 gms of Salt.
If this has to become 5 out of 12 parts, then the total amount of water should be = 180 ml.
So, the amount of water to be added is 80 ml.

Bookmark FeedBack

Q.66 [11594329]

Ramu has a box that has 71 candies. He is allowed to take out exactly 30 candies from the box or to add exactly 18 candies to it. If he is allowed to do this as many times as possible which of the following is the smallest number of candies that can be in the box at any point of time?

1 ○ 1

2 ○ 5

3 ○ 7

4 ○ 11

✖

Solution:
Correct Answer : 2
Your Answer : 1

Answer key/Solution

Use Options.
The number of candies is 71.
If X operations of removal and Y operations of addition are done, then the number of candies left = 71 – 30X + 18Y.
This has to be a number that leaves a remainder 2 when divided by 3. So options (1) and (3) are not possible.
Let us check if 5 is possible.
Then, 30X – 18Y = 66. (or) 5X – 3Y = 11. There is a solution at Y = 3 and X = 4.
Hence, 5 is the appropriate answer.

Bookmark FeedBack

Q.67 [11594329]

Two jars each contain the same number of marbles, and every marble is either blue or green. In Jar A the ratio of blue to green marbles is 9 : 1, and the ratio of blue to green marbles in Jar B is 8 : 1. There are 95 green marbles in all. The marbles from both the jars are emptied into Jar C. A marbel is randomly picked from Jar C. The probability that it is blue is:

1 ○ 1/2

2 ○ 4/5

3 ○ 17/36

4 ○ 161/180

Solution:
Correct Answer : 4

Answer key/Solution

Note: The "95 marbles" data is unnecessary.

The weighted average of the blue marbles = $\frac{9}{10}\times\frac{1}{2}+\frac{1}{2}\times\frac{8}{9}=\left(\frac{1}{2}\right)\left(\frac{9}{10}+\frac{8}{9}\right)=\frac{161}{180}$.

Bookmark FeedBack

Q.68 [11594329]

On the occasion of Diwali a shopkeeper is selling a gift hamper which has 5 boxes of sweets in it but is charging the list price of only 4 boxes. For every set of 4 such gift hampers bought by a customer, he is given 1 box of sweet as a free gift. If a customer buys 10 such gift hampers, then what is the percentage discount (nearest integer) availed by the customer?

✖

Solution:
Correct Answer : 23
Your Answer : 20

Answer key/Solution

Let the marked price of each box of sweets be Re. 1.
The customer buys 10 gift hampers.
∴ According to the question,
Total M.P. = 10 × 5 × 1 + 2 × 1 = 50 + 2 = Rs. 52
But the shopkeeper charged the list price of only 4 boxes for each gift hamper and gave 1 box free for every 4 such gift hamper.
∴ Total S.P. = 10 × 4 = Rs. 40

∴ Required discount% = $\frac{52-40}{50}\times 100 = \frac{12}{50}\times 100 \approx 23\%$.

Bookmark FeedBack

Q.69 [11594329]

A 10 × 6 rectangular sheet is folded once to form a 5 × 6 sheet. This sheet is folded now to make a 5 × 3 sheet. This new sheet is folded to make a 2.5 × 3 sized sheet, then a 2.5 × 1.5 sized sheet…and so on. The same process is done 10 times. Now all the paper is unfolded to bring it back to the original 10 × 6 sheet. What is the sum of the lengths of all the creases on the paper?

1 ○ $2^9 - 2^4$

2 ○ $2^{14} - 2^4$

3 ○ 2^9

4 ○ 2^{14}

Solution:
Correct Answer : 1

Answer key/Solution

The paper is folded 10 times, 5 times parallel to the length and 5 times parallel to the breadth.
The first fold parallel to the breadth introduces one crease of size 6 parallel to the breadth. The next fold parallel to the breadth introduces 2 more parallel to the width. The next one, 4 more...so on. The same is true with the folds parallel to the lengths.
So sum of all the folds parallel to the width are $(2^4 + 2^3 + 2^2 + 2^1 + 1) \times 6 = 6 \times (2^5 - 1)$
Similarly, sum of all the folds parallel to the length are $(2^4 + 2^3 + 2^2 + 2^1 + 1) \times 10 = 10 \times (2^5 - 1)$
Lengths of all the creases = $16 \times (2^5 - 1) = 2^4 (2^5 - 1)$.

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Q.70 [11594329]

If $\log_{1/10} 25x \geq 0$, then what is the least value of K for which $K^{2/3}.x$ is an integer?

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Solution:
Correct Answer : 125
Your Answer : 4

Answer key/Solution

$\log 25x \leq 0$
So $0 < 25x \leq 1$
$0 < x \leq \frac{1}{25}$
So smallest value of A so that A.x is an integer is 25.
$A = 5^2 = K^{2/3}$.
Solving for K, we get K = 125.

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Q.71 [11594329]

Sum of the first 7 terms of an AP is same as the sum of the first 18 terms. What is the 25th term of this AP if the sum of the first 24 terms is 100?

Solution:
Correct Answer : -100

Answer key/Solution

Sum of the first 7 terms = Sum of the first 18 terms.
So Sum of all the terms from the 8th term to the 18th term is 0.
The average of these terms is the 13th term.
This must be zero.
So sum of the 25 terms = 0.
$S_{24} + a_{25} = 0$.
So, $a_{25} = -100$.

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Q.72 [11594329]

The area of a circle with center O, is 64π. The side BC of an equilateral triangle ABC is the chord of this circle. The distance between the vertex A and the center of the circle O is $8\sqrt{3}$. The center of the circle lies outside the triangle and the vertex A lies outside the circle. What is the length of the side of the equilateral triangle?

1 ○ 16

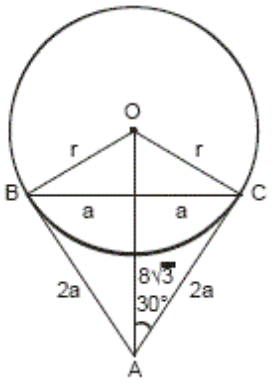
2 ○ 12

3 ○ 8

4 ○ 18

Solution:
Correct Answer : 3

Answer key/Solution



$\pi r^2 = 64\pi$
$\Rightarrow r^2 = 64$
Using cosine rule in triangle AOC,

$$\cos 30^\circ = \frac{(8\sqrt{3})^2 + 4a^2 - r^2}{2 \times 8\sqrt{3} \times 2a}$$

$$\Rightarrow r^2 = 4a^2 + 192 - \frac{\sqrt{3}}{2} \times 2 \times 2a \times 8\sqrt{3}$$

$\Rightarrow a^2 - 12a + 32 = 0$
$\Rightarrow (a - 8)(a - 4) = 0$
$\Rightarrow a = 4$ or 8
a = 8 is not possible as then the height of equilateral triangle will become $8\sqrt{3}$ which is equal to OA that means point O will lie on side BC but not outside the triangle.
$\therefore$ a = 4 is possible.
Hence, the length of the side of the equliateral triangle = 2 × 4 = 8.

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Q.73 [11594329]
A point P is situated inside a convex quadrilateral ABCD such that sum of the distances of the vertices from it is minimum. If PA, PB, PC, and PD are 6, 8, 12,and 16 respectively, then the maximum possible area of ABCD is

1 ○ 216

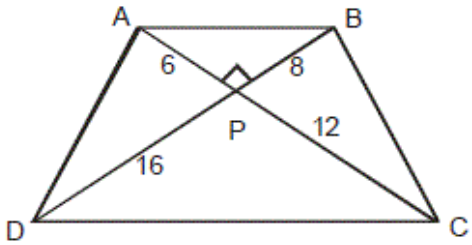
2 ○ 432

3 ○ 108

4 ○ None of these

Solution:
Correct Answer : 1

Answer key/Solution



If PA + PB + PC + PD has to be minimum, then APC, BPD have to be straight lines. So, P is the intersection of the diagonals.
If the lengths of the diagonals are 18 and 24, then the maximum area of the quadrilateral is when they are perpendicular to each other.

So, the maximum area = $\left(\frac{1}{2}\right)(18)(24) = 216$.

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Q.74 [11594329]

A shopping complex has 3 mobile stores A, B and C which sell mobiles of only two brands Oneplus and Apple. The ratio of number of Oneplus mobiles to Apple mobiles in stores A, B and C are in ratio 3 : 4, 4 : 5 and 8 : 3 respectively. The ratio of number of Oneplus mobiles to Apple mobiles in store A and B combined is 7 : 9 and that in store B and C combined is 5 : 2. Then which of the following could be the total number of mobiles in these three stores?

1 ○ 340

2 ○ 406

3 ○ 390

4 ○ 410

Solution:
Correct Answer : 2

Answer key/Solution

Let the number of Oneplus mobiles and Apple mobiles in store A be 3x and 4x, that in store B be 4y and 5y and that in store C be 8z and 3z respectively.
According to the question,

$$\frac{3x+4y}{4x+5y}=\frac{7}{9}$$

$\Rightarrow x = y$

And $\frac{4y+8z}{5y+3z}=\frac{5}{2}$

$\Rightarrow z = 17y$
So, the total number of mobiles in these three stores
$= 7x + 9y + 11z$
$= 7y + 9y + 11 \times 17y$
$= 203y$
$\therefore$ Among the options, 406 could be the possible answer.

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Q.75 [11594329]
Two candles of equal length are lit at noon. One candle takes 10 hours to completely burn while the other takes 5 hours to completely burn. At what time will the slower burning candle be exactly twice as long as the faster burning one?

1 ○ 3.5 hours

2 ○ 3.33 hours

3 ○ 2.5 hours

4 ○ 2 hours

✖

Solution:
Correct Answer : 2
Your Answer : 3

Answer key/Solution

Let the lengths of both the candles be 10 cm. Speeds of burning are 1 cm/hr and 2 cm/hr.
If T is the time after which the length of the candle that burns slower is twice the length of the faster candle, then 2 (10 – 2 T) = 10 – T. So 10 = 3T.
Hence, T = 3.33 hours.

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